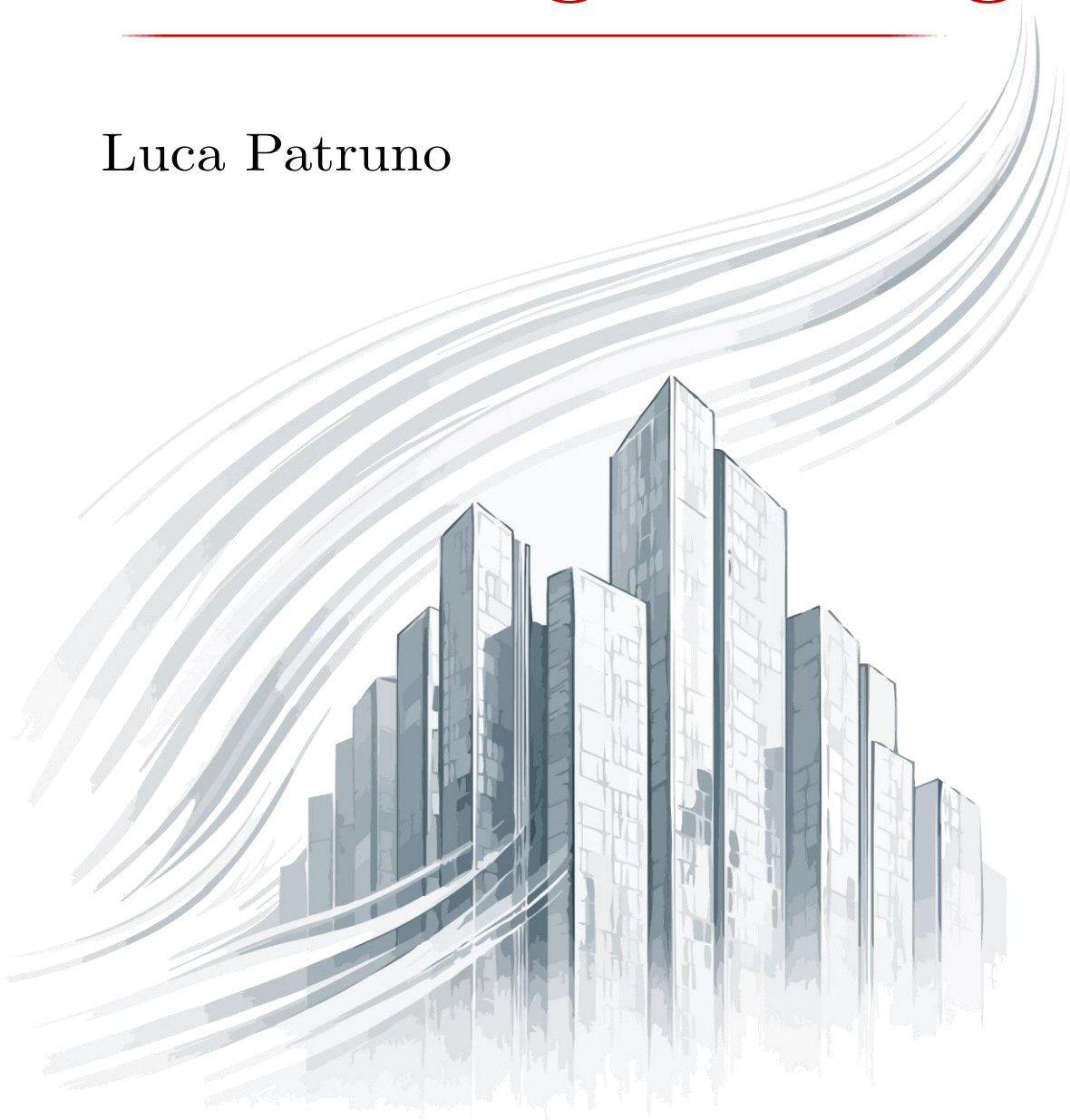


Essential Concepts in

Computational Wind Engineering

Luca Patruno



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Computational
Wind Engineering

First edition - Revision II

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First published: February 10, 2026
Revised: October 7, 2026

To my son, Adriano,
my wife, Jeng Yi,
Benedetta, my soul,
my parents Anna Rita and Roberto,
Antonella and Ivan, my legacy.

Preface

Wind is the most ethereal of the environmental forces, yet it can be relentless and unforgiving at times. It is the most capricious of the natural elements: destructive when furious, gentle when calm, elusive in its temper.

Despite its strength, its ethereal nature seems powerless beside the firmness of stones and bricks used to build man-made structures for most of human history. In particular, ancient structures are usually characterised by high self-weight, with gravity loads being necessary to ensure stability and predominant with respect to all other applied loads. An exception to this is represented by earthquake-induced loads, which have in fact significantly contributed to their destruction. However, throughout history, technological progress in construction and material technologies have led to a constant decrease in the structural weight.

A sketch of the evolution of structural weight throughout history is shown in Fig. 1. It can be noted that the first structures which approached the modern ones in terms of lightness are timber structures that appeared in Asia starting from the 5th century, predating others by more than a millennium. However, in the last century, the use of thin reinforced concrete shells first, and of tensile structures later, have led to structural systems which are lighter than 1 kN/m^2 [1].

The transition from heavy, gravity-stabilised, structural systems to light typologies is a paradigmatic one, which changes the relative importance of permanent and accidental actions. Self-weight becomes comparable or dominated by accidental loads, with wind often occupying a prominent position among them.

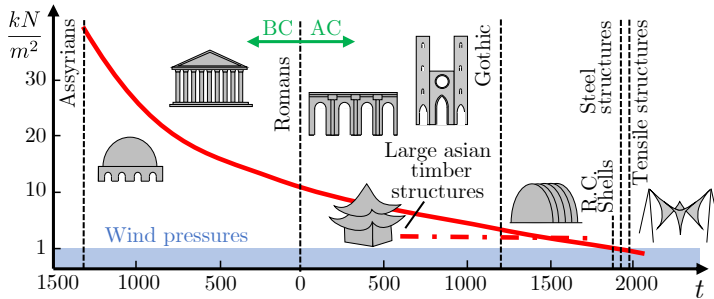


Figure 1: Qualitative evolution of the structural weight throughout history (axes not to scale). *Image redrawn and integrated from [1], courtesy of M. Majowiecki.*

This book is the result of the author's efforts in preparing the Computational Wind Engineering course at the University of Bologna, dedicated to master's students in Civil Engineering. Its aim is to give readers a strong background in the subject, supporting practical application and facilitating further in-depth study.

In particular, Wind Engineering is the discipline which studies the interaction between wind and the built environment, together with the structures contained within it. Its origins can be traced back to ancient times, with its roots grounded in meteorological observation and shipbuilding [2]. It should be observed that the contribution of shipbuilding first, and Naval Engineering later, to the modern Wind Engineering is not coincidental. In fact, the whole tradition of modern light structures moved fundamental steps in that field, in which the need for high structural performance together with low weights was literally dictated by the need to stay afloat. A clear example is the development of tensile-structures, the lightest between the structural typologies, which originated from the development of sails.

Proceeding in time, Aeronautical Engineering branched out from Naval Engineering, giving rise to another field that has greatly contributed to modern Wind Engineering. For instance, Gustave Eiffel, one of the founding fathers of wind tunnel testing, inaugurated his first wind tunnel in 1909 at the base of his eponymous tower, taking advantage of its powerful electrical system. The facility was devoted to the testing of aeronautical components, highlighting that the design of light structures, whether towers or aeroplanes, was at that time largely transversal to specific applications.

Scientific and technological advancements have greatly progressed since Eiffel's times, sharpening the boundaries between disciplines. Such subdivisions sometimes mark specific competence niches but, more often, denote a loss of awareness regarding common cultural references, whose integrity and complementarity were once manifest. Wind Engineering stays at a point where the aforementioned boundaries meet, still in a privileged position to appreciate the intimate connections existing between them.

Over the past decades, this already rich body of knowledge has grown and further diversified due to the increasing importance of Computational Wind Engineering, which studies problems typical of Wind Engineering relying on numerical simulations [3]. An overview of the historical development of the subject can be found in [4]. At first sight, the difference between "classical" and Computational Wind Engineering is mostly operational, being limited to the adopted investigation facility. However, the advent of Computational Wind Engineering is a much deeper transformation and it is characterised by interconnected opportunities and dangers [5].

Among the opportunities, there is undoubtedly a deeper (re)connection of the discipline with some of its roots and offspring of different engineering branches. In fact, a basic knowledge of Continuum Mechanics, Fluid Dynamics, Turbulence Modelling and Numerical Analysis is an essential prerequisite for a fruitful approach to the subject.

Among the dangers, one of the most significant is paradoxically represented by its increasing accessibility. The need for wind tunnels, present in a few specialised centres, has always been in itself an assurance of appropriate training for the personnel operating in the field. Numerical simulations may free operators from the need for dedicated infrastructures, but do not exempt them from acquiring the technical background required to confront the most capricious of the environmental forces.

In effect, the cultural journey required to obtain a fairly clear picture of the subject is entangled and, at first, repulsive. The problem lies in the close interconnection between the subjects which contribute to the overall picture, which are usually much vaster than the needs encountered in the usual practice of Computational Wind Engineering. As a result, when studying on the classical texts used in each of the aforementioned disciplines, the newcomer is exposed to a myriad of often disorienting stimuli. The consequence is that the learning process is slowed down by an over-exposure to details, much before a sufficiently clear overall picture of the matter is matured.

From all points of view, this book is nothing more than a timid summary of the aforementioned disciplines that collectively underpin Computational Wind Engineering, intended to foster an organic understanding of their interrelations.

The genesis of the book structure is itself eloquent in this regard. While preparing the Computational Wind Engineering course material, the author soon realised that it would have been impossible to undertake the aforementioned entangled journey within the available time. The author thus decided, not without hesitation, to write the extended notes of the course, attempting to straighten the path as much as possible. It was decided to limit the material to ten chapters which, by experience, is a good number for the main topics of a one-semester course.

Given the available time, each chapter was limited to approximately thirty pages, figures included, with only few exceptions. The final chapter, which covers application examples, was later added to provide applied examples that bridge theory and practice. Finally, the material was subdivided into four parts to highlight the progression from fundamentals to applications.

As it can be imagined, completeness and depth were not considered as objectives in the writing process, which was instead characterised by drastic cuts and deliberate omissions. The most notable example is the absence of considerations related to Structural Dynamics involving concepts more advanced than the simple oscillator. Conceptual gaps, however, have been carefully avoided. Whenever simplifications or omissions are retained, they are explicitly stated and clearly signaled to the reader.

Overall, half of the chapters are dedicated to topics which might be directly ascribed to Wind Engineering, while the other half is dedicated to topics which, collectively, might be regarded as Computational Fluid Dynamics. Despite the aforementioned limitations, the author believes that the resulting material meets most of the initial objectives. It offers a self-contained and consistent overview of the essential concepts underlying Computational Wind Engineering, useful both in itself and as a foundation for approaching more advanced classical textbooks, e.g. [6, 7, 8, 9, 10].

The order of presentation follows the lecture schedule and does not necessarily represent the most logical thematic subdivision. Vertical bars alongside the text highlight the most important passages, allowing the reader to follow only these marked sections for a concise or focused reading.

A special mention should be made regarding references. Also in this regard completeness was not sought. The book is didactic in nature, so that preference has been given to books rather than scientific papers. In particular, books, including those cited above, can be considered as suggestions for further reading or consultation. Scientific papers are either fundamental references or are used as sources for illustrations and examples. In particular, when papers are from the author, they surely fall within the second of such categories.

As a final remark, it is noted that in a few circumstances the book indulges in details that might have been omitted to obtain a more direct and evenly paced narration. Such passages are almost always propaedeutic to further developments. In a few instances, however, the author chose not to spare the reader some digressions, considering that further omissions would have obfuscated the inherent beauty of the subject, whose appreciation may, in itself, be a worthy reason to embark on this journey.

Bologna, January 6th 2026

Artificial intelligence was used to assist with language editing and the creation of the cover image. Content and conclusions are the author's own.

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Notation

To begin, we want to warn the reader that the notation used in this book is not completely standard. In particular, for Continuum Mechanics we have three main notations in use: *vector calculus notation* (standard for Computational Fluid Dynamics), *operator notation* (standard for Structural Mechanics), *index notation* (more abstract). Each notation has its strengths and limitations, representing a compromise between historical development, expressivity and ease of use.

Readers who have not been deeply exposed to any of the aforementioned notations can safely proceed to Chapter 1, as explanations are gradually provided in the text. Others might want to clarify this aspect before reading further.

In particular, we attempt to retain the expressivity of *vector calculus notation* and the operational effectiveness of *operator notation* using minimal tweaks. We acknowledge that, to a certain extent, this is possible only within the limited scope of the present book. The author hopes to help the most, while discontending others only to an acceptable level.

The following list of symbols and identities might be used in the following to avoid misunderstandings. We take as example the velocity vector, $\mathbf{u}$, that will recur in our discussion

u	scalar
$\mathbf{u} = [u, v, w]^T$	vector (column)
∇	$[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]^T$ (column)
∇u	$[\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}]^T$ (gradient, column)
$\nabla \cdot \mathbf{u} = \nabla^T \mathbf{u} = \nabla(u)$	$[\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}]$ (gradient, row)
$\nabla \cdot \mathbf{u}$	$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$ (divergence)
$\nabla \cdot \nabla(u) = \nabla \cdot \nabla \mathbf{u} = \nabla^2(u)$	$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ (Laplacian).

Essentially, ∇ is a column vector which contains a differential operator. The dot product indicates scalar-vector multiplication (implying transposition of ∇) or scalar product between vectors, depending on context (despite the orientations of the vectors). The symbol $\nabla(u)$ indicates ∇ operating on u , resulting in the gradient as a row. The list above is not complete, but other operations follow usual conventions. As usual, the divergence of a second-order tensor is always considered to be a column vector.

Part I

Wind and Flow Fundamentals

Chapter 1

The atmospheric boundary layer

Structures are located within the lowest portion of the atmosphere, where the presence of the ground exerts a strong influence on the motion of air. In this chapter, we set the stage for our investigation by characterising wind through appropriate metrics. Doing so requires the introduction of several fundamental concepts. These concepts will recur throughout the book and will gradually clarify as they are repeatedly used...

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1.1 Davenport’s wind-load chain

As is well known from everyday experience, wind has a random character, meaning that its evolution cannot be predicted without a great deal of uncertainty. Specifically, this simple observation can be applied to two slightly different situations: (i) it is difficult to predict the average wind speed and direction between 2.00 PM and 3.00 PM the day after tomorrow and, (ii) given that there is strong wind now, predict the instantaneous wind speed and direction in 2 min time. The two problems are indeed similar, but they take place at different scales [11].

The first case concerns the prediction, although uncertain, of the evolution of the meteorological conditions at the *macro-scale* (planetary scale, say 1000 km or 3 days and above) and at the *meso-scale* (say 10-100 km or 1 h and above, up to the macro-scale).

Conversely, the second case concerns the prediction of the local wind condition at the *micro-scale*, which includes all the length- and time-scales below the meso-scale. This is the scale at which, for instance, we might consider the presence of the surroundings like buildings and trees.

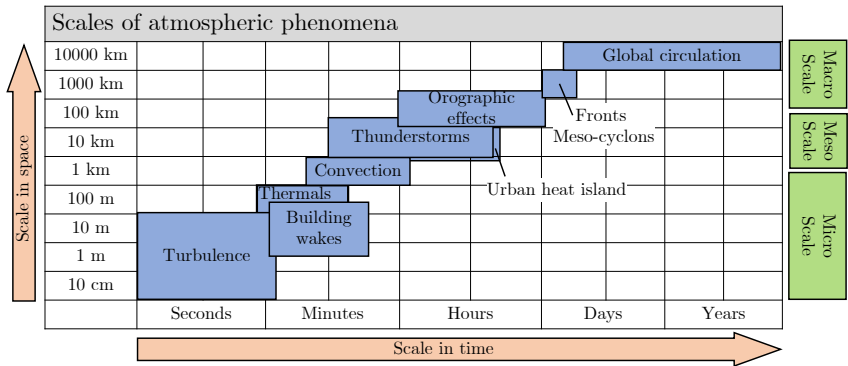


Figure 1.1: Length-scales and time-scales of meteorological phenomena, after [11].

A graphical representation of the meteorological scales and the phenomena taking place in each of them is shown in Fig. 1.1. Van der Hoven in 1957 provided a clear representation of the different time-scales present in wind [12]. In particular, he measured the energy contained by wind at different time-scales, producing the graph reported in Fig. 1.2. As it will be later clarified (see Sec. 1.7.2), this plot is called *spectrum* and, for wind, it takes the name of Van der Hoven wind spectrum.

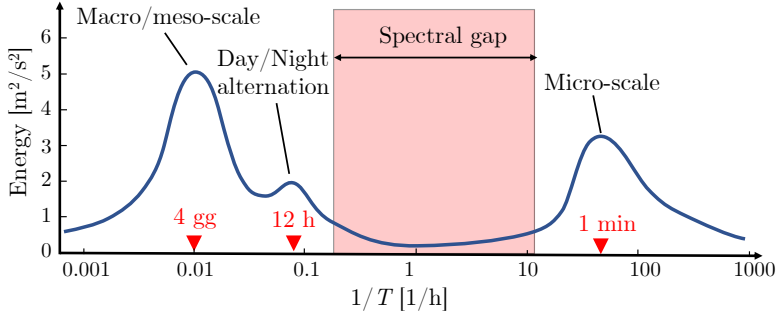


Figure 1.2: Van der Hoven wind spectrum, after [12].

The Van der Hoven wind spectrum clearly shows a peak corresponding to the macro-meso scale (approximately 4 days, the characteristic evolution time of atmospheric perturbations) and at the micro-scale, say 1 min.

It can be seen that a gap, characterised by low energy, is present between the macro/meso-scale and the micro-scale, denoted as the *spectral gap*. Essentially, meteorological phenomena at the macro/meso-scale are well-distinguishable from local effects due to the particular topography and obstacles encountered by wind.

The existence of the spectral gap opens the possibility to study the two scales separately, and it is at the base of the so-called Davenport's chain of wind loading, depicted in Fig. 1.3. The Davenport's chain represents the outline of any Wind Engineering study.

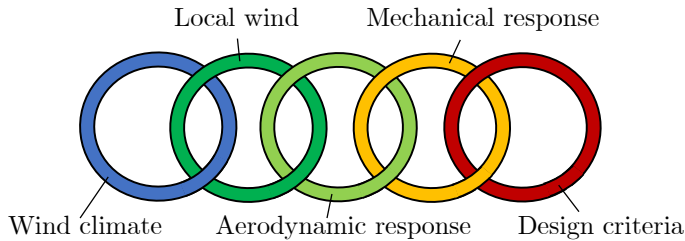


Figure 1.3: Davenport's wind load chain, after [13].

According to Davenport, the study should be subdivided into the five main steps reported below

- i) **Wind climate** - it is necessary to characterise the maximum velocity which can be expected on site due to macro/meso-meteorological structures;

- ii) **Local wind** - it is necessary to characterise the effects of local conditions on the macro/mesosopic wind speed (e.g. height from the ground, site elevation, orography, terrain roughness and obstacles);
- iii) **Aerodynamic response** - each structure has a particular aerodynamic response to the local wind conditions due to its shape;
- iv) **Mechanical response** - each structure has a particular mechanical response as a consequence of aerodynamic loading;
- v) **Design criteria** - once the structural response is estimated, it is necessary to compare it to appropriate design criteria to assess its acceptability.

In the following, we begin to delve into the aspects that define each link of Davenport's chain.

1.2 Global circulation

The global circulation, i.e. the movement of air masses at the macro/meso-scale, is essentially powered by temperature differences induced by uneven solar radiation. With a great deal of simplification, the air motion at the planetary scale self-organises in *convective cells*, which are depicted in Fig. 1.4 (a). Moving from the equator to the poles, we encounter the Hadley, the Ferrel (mid-latitude) and polar cells. These cells extend up to the troposphere, at around 10-15 km altitude, i.e. the cruise altitude for aeroplanes.

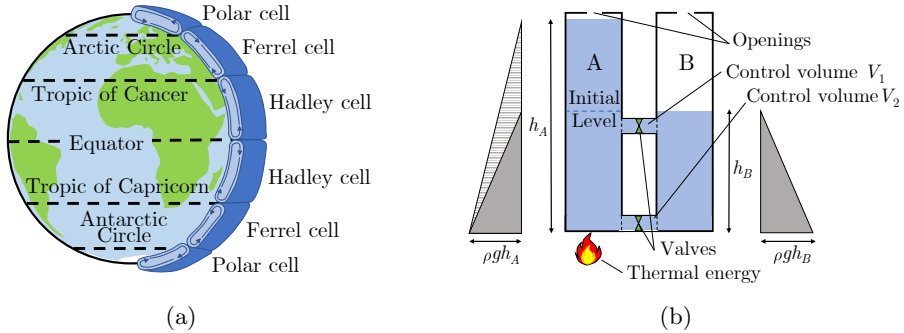


Figure 1.4: Global circulation: (a) convective cells at the planetary scale and (b) schematic of a convective cell.

An explanation providing some intuition of the inner working mechanisms of a convective cell can be found in [14]. We consider a device composed of two tanks placed vertically with an opening on the top, linked by two horizontal pipes, see Fig. 1.4 (b). Two valves are present in the horizontal pipes, which are initially closed. At the beginning the two tanks are filled with the same water height $h = h_A = h_B$, so that the pressure profile is null at the water level and equal to ρgh at the bottom of the tanks, following a linear variation with height. This pressure profile is shown as solid in Fig. 1.4 (b).

Looking at the control volume V_1 (surrounded by the top horizontal pipe), we observe that equal pressure is applied on both sides, so that the control volume is in static equilibrium. The same applies to the control volume V_2 , surrounded by the bottom horizontal pipe. As a result, if we open the valves, the fluid does not move.

We now supply thermal energy at the bottom of tank A. The fluid expands while keeping its weight, i.e. density decreases. The pressure profile established after the expansion is shown as a dashed area in the figure. It can be seen that, if we open the bottom valve, no flow occurs, as the control volume V_2 is still in equilibrium.

However, the control volume V_1 is not in equilibrium (actually it is in equilibrium due to the reaction provided by the valve while it is closed), so that, if we open the valve, a flow is observed, until the pressure applied on both sides of V_1 equalises. This leads to an increase in the water height in tank B, leading to an increase of pressure at its bottom. Due to the increase in pressure, V_2 is no longer in equilibrium, so that when the valve is opened, a flow from tank B to A is observed. Keeping both valves open, the flow continues as long as there is a temperature difference between tank A and B.

This model explains well the motion in the polar and Hadley cells (the latter one being located between the equator and the tropics), while the Ferrel cell (mid-latitude) moves in the opposite direction to the aforementioned mechanism. This is due to the fact that the Ferrel cell is an indirect circulation cell, whose motion is sustained by the presence of the adjacent Hadley and polar cells.

In summary, the driving force behind wind motion at the planetary scale is represented by pressure gradients, which are induced directly or indirectly by temperature gradients. Wind flows from high- to low-pressure regions, i.e. against the pressure gradient.

Another key aspect to be observed is that wind does not flow along meridians, as it should be expected if we consider only the effects described

above. This is due to the presence of the Coriolis force.

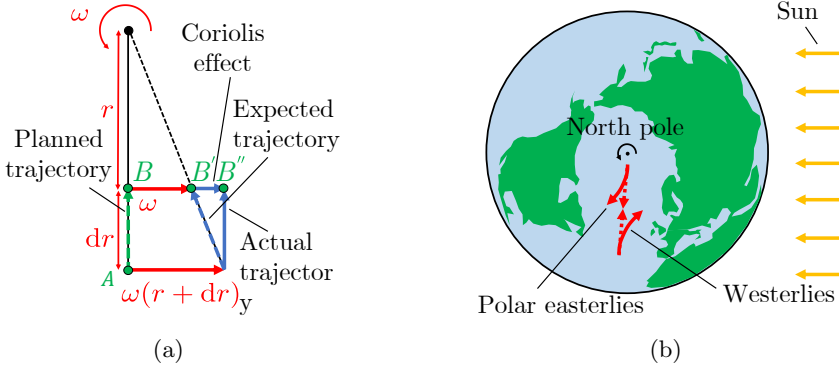


Figure 1.5: Coriolis force: (a) schematic explanation of its working mechanism and (b) effect on global circulation.

It is well known that the Coriolis force is an apparent force, which manifests itself when a body moves on the surface of a spinning object. Assume to adopt a reference frame fixed to the spinning object, so we denote it as spinning reference frame. The acceleration induced by the Coriolis force in the spinning reference frame, $\mathbf{a}_c$, can be calculated as

$$\mathbf{a}_c = 2\mathbf{u} \times \boldsymbol{\omega}, \quad (1.1)$$

where $\mathbf{u}$ is the velocity vector of the moving object in the spinning reference frame, $\boldsymbol{\omega}$ is the angular velocity vector of the spinning object and $\times$ is the cross product.

To obtain an intuitive grasp on its working mechanism, let us consider Fig. 1.5 (a). A body is moving on a spinning disk, going from the periphery to the centre, where the rotation axis is placed. At time t , the body is found in A and moves toward point B.

Viewed from an external (i.e. inertial) reference frame, at point A the velocity component in the tangential direction is $\omega(r + dr)$. The body moves in the direction of B without receiving any force. According to the expected trajectory, at time $t + dt$ the body should be found in B', i.e. the new position of B. However, due to the difference in tangential velocities on the disk at different distances from the spinning axis, the body is found in B''. Ignoring the fact that the effect is caused by the rotation of the disk, the observer in the spinning reference frame attributes this result to the presence of a force, causing an acceleration that deflects the body from the planned trajectory.

In particular, we can equate the position difference caused by the Coriolis effect to that caused by an acceleration obtaining

$$\omega(r + dr)dt - \omega r dt = \frac{1}{2}a_c dt^2, \quad (1.2)$$

leading to

$$a_c = 2\omega \frac{dr}{dt} = 2\omega u_r, \quad (1.3)$$

which is the particularisation of Eq. (1.1) to the present case.

Applying the same considerations to the global circulation we obtain, for instance, the results reported in Fig. 1.5 (b). At mid-latitudes in the northern hemisphere, winds are deflected west, leading to the so-called westerlies. The polar easterlies result from similar effects.

Overall, it can be said that the global circulation is driven by the sun, and it is strongly influenced by the rotation of the Earth. Pressure gradients are the cause of the motion and Coriolis forces deflect the trajectory of air masses. Winds arising from the balance of pressure gradients and Coriolis forces (and to a lesser extent centrifugal forces) are denoted as *geostrophic winds*.

In addition to the mechanism above, numerous factors contribute to shaping the wind climate, giving rise to a variety of meteorological phenomena. For instance, large water bodies heat up and cool down more slowly than continental masses, leading to seasonal circulations such as monsoons. Cyclones (or typhoons, depending on the region) develop primarily from the release of latent heat during moist convection over warm ocean waters. Other phenomena lead to more localised effects such as sea breezes, foehn, downbursts, tornadoes etc. The importance of such winds, which can be significant, greatly depends on the geographical location and it is outside the scope of the present discussion. The interested reader can find additional details in the dedicated literature, see for instance [15].

For our purposes, we concentrate on *synoptic winds*, those taking place at the macro-scale and the largest structures at the meso-scale. We also assume that wind speed has been measured over a sufficiently long period at each location of interest. This allows statistical analyses to estimate wind velocities that are expected to occur with sufficiently low probability over the service life of a structure. However, looking closely at the sentence above and wishing to put it into practice, a major question arises. How should we define wind velocity?

1.3 The atmospheric boundary layer

To answer the question, let us consider a thought experiment. We place three anemometers at different heights above the ground at our site of interest, say P_1 , P_2 and P_3 , and record the wind velocity time histories shown in Fig. 1.6. This is a *time-domain* representation of our observation. We immediately observe that the wind speed varies in time in an unpredictable manner, so that capturing it with a single value is a difficult task. However, recalling the spectral gap, we know that macro/meso-scale meteorological structures have a characteristic time of a few hours at least, while structures belonging to the micro-scale have characteristic time shorter than, say, 10 min. In other words, the rapid fluctuations we see in the wind speed are micro-scale effects.

It follows that by time-averaging the wind speed for 1 h or 10 min, fluctuations due to micro-scale effects are removed, leading to a velocity representative of the one at the macro/meso-scale. Additionally, averaging over 1 h or 10 min should not differ much, as such intervals both fall in the spectral gap, in which no major source of velocity fluctuation is expected. This is in fact the case, with the velocity $u(t)$ averaged over 10 min, $U_{10\min}$, being such that $U_{10\min}/U_{1h} \approx 1.05 - 1.10$, depending on the local climate. Evaluating the wind speed at the macro/meso-scale using averaging times of 10 min or 1 h is the usual approach followed by design codes.

In the following, for illustrative purposes, we describe the wind characterisation adopted in Eurocode [16]. However, while specific aspects may vary when adopting other standards, the main ideas presented below are of general validity.

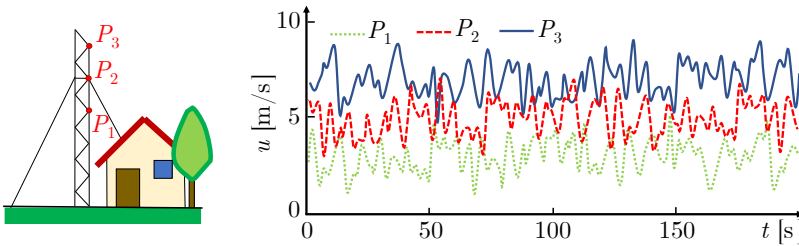


Figure 1.6: Wind time histories measured at different distances from the ground.

In particular, looking again to Fig. 1.6, we observe that velocity increases with height. This is because the ground exerts friction on the overlying moving air masses, opposing their motion. This causes the wind to “slow

down” close to the ground. This friction is actually a *shear stress*, as it is transmitted in a direction orthogonal to the ground normal. The result is that the variation of the wind speed with altitude assumes a characteristic logarithmic shape. The rougher the ground, the higher is the shear stress acting on the moving air, leading to the so-called *wind profiles*, depicted in Fig. 1.7. The zone near the ground in which the flow is affected by ground friction is denoted as Atmospheric Boundary Layer, ABL.

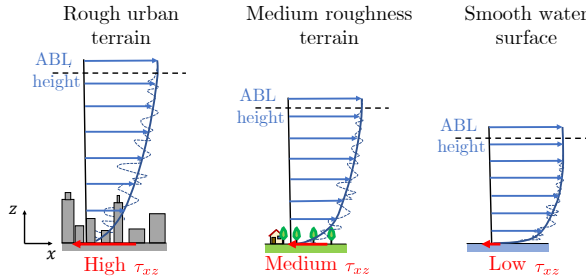


Figure 1.7: Wind profiles for terrains with different roughness, image after [17].

We are now in a position to start answering our initial question. A quantification of the wind velocity requires to separate macro/meso-scale effects from micro-scale ones. The macro/meso-scale wind characterisation requires to consider the $U_{10\min}(z)$ profile. However, $U_{10\min}(z)$ depends on the *terrain roughness*, which is a micro-scale feature.

In practice, the terrain roughness is usually evaluated visually, according to the classification in Fig. 1.8 or similar ones. This leads to the selection of a *terrain category*.

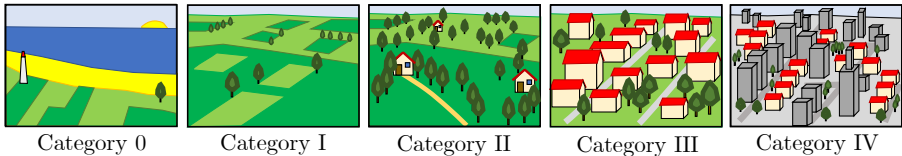


Figure 1.8: Terrain roughness categories according to Eurocode [16]. Drawings are redrawn by the author and are not the official ones reported in the standard.

Then, the $U_{10\min}(z)$ profile is evaluated according to the formula

$$U_{10\min}(z) = U_b k_r \ln(z/z_0), \quad (1.4)$$

where U_b is the *base velocity* detailed below, k_r is the *terrain factor*, and z_0 is the *aerodynamic roughness*.

For design purposes, the minimum allowable velocity is taken at z_{min} reported in Fig. 1.9, but we will not dwell on such aspects here.

The values of k_r and z_0 can be obtained from the table reported in Fig. 1.9, as a function of the terrain category. In particular, the aerodynamic roughness, z_0 , can be related to the characteristic size of the average obstacle found on the terrain, expressed through the *sand-grain roughness*, k_{sd} , and, as a rule of thumb, $k_{sd} \approx 30z_0$ [18]. For instance, as reported in Fig. 1.9, a suburban area (terrain category III) which might have obstacles of typical size in the order of 10 m is assigned $z_0 = 0.3$ m.

The terrain factor, k_r , can be calculated according to

$$k_r = 0.19 \left(\frac{z_0}{z_{0,II}} \right)^{0.07}, \quad (1.5)$$

where $z_{0,II} = 0.05$ m is the z_0 for the category II terrain.

Terrain category		z_0 [m]	z_{min} [m]	k_r [-]
0	Sea or coastal area exposed to the open sea	0.003	1.0	0.16
I	Lakes of flat and horizontal area with negligible vegetation and without obstacles	0.01	1.0	0.17
II	Area with low vegetation such as grass and isolated obstacles (trees, buildings) with separation of at least 20 obstacles heights	0.05	2.0	0.19
III	Area with regular cover of vegetation or buildings or with isolated obstacles with separation of maximum 20 obstacles heights (such as villages, suburban terrain, permanent forest)	0.3	5.0	0.22
IV	Area in which at least 15% of the surface is covered with buildings and their average height exceeds 15 m	1.0	10.0	0.24

Figure 1.9: Profile parameters values for different terrain roughness categories according to Eurocode [16].

The logarithmic profile reported in Eq. (1.4) is a general one, representative of the increase of wind speed with height in most conditions. For design purposes, however, we are interested in obtaining the most severe wind speed distribution that the structure might reasonably experience in its service life. Therefore, we select the *base velocity*, U_b , which multiplies the profile, to be representative of an appropriately high value.

In particular, the base velocity U_b , by definition, represents the 10 min averaged velocity for a terrain category II (typically found around airports), measured at 10 m height, with a return period of 50 years. Note in fact that by substituting the values of z_0 and k_r of the category II terrain, into Eq. (1.4) we obtain $U_{10min}(z) = U_b$ for $z = 10$ m while, at the same height, lower values are obtained for rougher terrains and higher values for smoother ones.

The value of U_b is provided in national building codes on the basis of statistical studies relying on Extreme Value Analysis (further details given in Sec. 9.4). In the case of Italy, for instance, the U_b value for sites at low heights above sea level is given in Fig. 1.10, and corrections are suggested in order to account for the site altitude. We note that the base velocity often shows jumps at national borders, as a result of the simplifications made to group areas characterised by similar values.

In summary, for design purposes, standards usually condense all the information required to characterise the wind climate at the macro/meso-scale at a given location in a single parameter, the base velocity (or equivalent), useful to represent extreme events. The effect of the ground on the time-averaged wind profile, which leads to $U_{10\min}(z)$ according to Eq. (1.4), is assumed to be dependent only on the terrain roughness, estimated relying on the terrain categories depicted in Fig. 1.8.

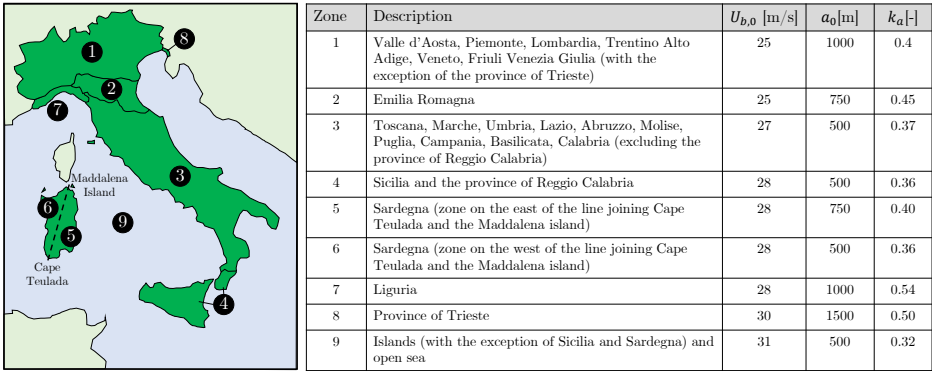


Figure 1.10: Base velocity map for Italy according to [17].

However, as already discussed, characterising $U_{10\min}(z)$ captures only a small portion of what we observed in our thought experiment. In fact, the velocity fluctuates continuously due to micro-scale effects. To provide a more refined description, it is therefore necessary to account for these fluctuations which, in turn, requires the introduction of a statistical framework into our discussion. We warn the reader that, in the following, the exposition becomes dense at times. A first reading may therefore skip some details, which can be revisited on a second reading for a more complete understanding. The importance of the subject justifies this layered approach and, upon closer inspection, the mathematics in this chapter is elementary, although it requires careful reading.

1.4 Random variables

1.4.1 Statistical distributions

The velocity in the ABL is not deterministic and should be, thus, analysed using a probabilistic approach. In particular, let us begin by measuring the instantaneous wind velocity (in a predefined direction) at a point in the ABL at J discrete times, say every 1 s. We collect results in a vector such that

$$\hat{\mathbf{u}} = [\hat{u}(t_1), \hat{u}(t_2), \hat{u}(t_3), \dots, \hat{u}(t_J)], \quad (1.6)$$

which we denote as *sample*. By plotting it as a function of time, it is possible to visualise a time history of the velocity at the point of interest, similarly to Fig. 1.6. It is, however, also possible to build the empirical Cumulative Distribution Function, CDF, $\hat{F}(u)$

$$\hat{F}(u) = \frac{\#(\hat{\mathbf{u}} < u)}{\#(\hat{\mathbf{u}})}, \quad (1.7)$$

where $\#$ denotes the cardinality. Note that, by construction, $\hat{F}(u)$ has a null value on the left of the lowest measured value and is equal to unity on the right of the highest measured value, see Fig. 1.11 (a).

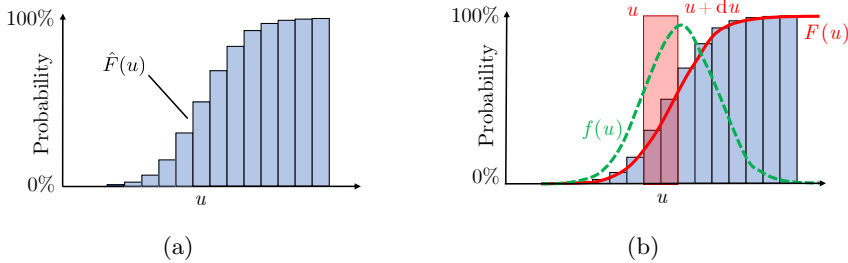


Figure 1.11: Statistical distributions: (a) sample CDF and (b) real CDF and PDF.

As the cardinality of $\hat{\mathbf{u}}$ increases, it is possible to note that $\hat{F}(u)$, which is sensitive to the particular realisation of $\hat{\mathbf{u}}$ for small samples, becomes stable and, in the limit of $\# \hat{\mathbf{u}} \rightarrow \infty$, it approaches the CDF of the *random variable* u , $F(u)$. From this perspective, each element of the sample $\hat{\mathbf{u}}$ is a particular extraction from the underlying random variable u . This is a *statistical* representation of our observation.

Note that the function $F(u)$ should not be confused with its empirical counterpart $\hat{F}(u)$, which is always obtained using a sample of limited cardinality, see Fig. 1.11 (b).

Once the CDF is known, it is often useful to evaluate how many realisations of u we can expect to lie in a certain small interval of values. In particular, we would like to quantify the number of samples falling between the values u and $u + du$, which is $\#(\hat{\mathbf{u}} < (u + du)) - \#(\hat{\mathbf{u}} < u)$. By using Eq. (1.7), this number can be written as

$$\begin{aligned}\#(\hat{\mathbf{u}} < (u + du)) - \#(\hat{\mathbf{u}} < u) &= F(u + du)\#(\hat{\mathbf{u}}) - F(u)\#(\hat{\mathbf{u}}) \\ &= [F(u + du) - F(u)]\#(\hat{\mathbf{u}}).\end{aligned}\quad (1.8)$$

As du has been supposed to be small, we can use a Taylor series to express $F(u + du)$. To first-order approximation, Eq. (1.8) can be rewritten as

$$\begin{aligned}\#(\hat{\mathbf{u}} < (u + du)) - \#(\hat{\mathbf{u}} < u) &= \left[F(u) + \frac{dF}{du}du - F(u) \right] \#(\hat{\mathbf{u}}) \\ &= f(u)du\#(\hat{\mathbf{u}}),\end{aligned}\quad (1.9)$$

where $f(u) = \frac{dF}{du}$ is the Probability Density Function, PDF. Note that knowing the CDF or the PDF provides a complete description regarding the statistical distribution of the random variable u .

1.4.2 Statistical moments

An alternative approach to characterise a random variable is to characterise its *statistical moments*. To understand such description, we first need to recall the similarity between the *scalar product* between two vectors and the $\mathcal{L}_2$ *inner product* between two functions.

In particular, let us consider two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Their scalar product reads

$$\mathbf{v}_1^T \mathbf{v}_2 = \mathbf{v}_1^T \frac{\mathbf{v}_2}{|\mathbf{v}_2|} |\mathbf{v}_2| = \mathbf{v}_1^T \mathbf{n}_2 |\mathbf{v}_2|, \quad (1.10)$$

where $\mathbf{n}_2 = \mathbf{v}_2/|\mathbf{v}_2|$ is the unit vector oriented in the direction of $\mathbf{v}_2$, $|\cdot|$ indicates the vector norm and the superscript T denotes transposition. The product $\mathbf{v}_1^T \mathbf{n}_2$ represents the component of $\mathbf{v}_1$ in the direction pointed by $\mathbf{v}_2$, while $|\mathbf{v}_2|$ can be seen as a scaling coefficient. Despite such scaling coefficient, the scalar product calculates projections between elements of a vector space.

Functions can be seen as infinite-dimensional vectors collecting the function values at all points, and the scalar product can be generalised to the $\mathcal{L}_2$ inner product.

In particular, considering functions $v_1(u)$ and $v_2(u)$, we can calculate the $\mathcal{L}_2$ inner product between $v_1(u)$ and $v_2(u)$ as

$$\mathcal{L}_2(v_1(u), v_2(u)) = \int_L v_1(u)v_2(u)du, \quad (1.11)$$

where L is the domain of interest where $v_1(u)$ and $v_2(u)$ are defined.

Notice that the $\mathcal{L}_2$ inner product expressed by the right-hand side of Eq. (1.11) is analogous to the scalar product expressed by the left-hand side of Eq. (1.10) since

$$\int_L v_1(u)v_2(u)du \approx \sum_{j=1}^J v_1(u_j)v_2(u_j)\Delta u = \mathbf{v}_1^T \mathbf{v}_2 \Delta u, \quad (1.12)$$

where the integral has been approximated with a finite summation. In particular, functions $v_1(u)$ and $v_2(u)$ are evaluated on a discrete number of points u_j , evenly separated in L by the distance Δu , so that $\mathbf{v}_1 = [v_1(u_1), v_1(u_2), \dots, v_1(u_J)]$ and, similarly, for $\mathbf{v}_2$. The approximation becomes an equality for vanishing Δu and, thus, J tending to infinity.

In this sense, the $\mathcal{L}_2$ inner product plays the same role in function spaces as the scalar product does in finite-dimensional vector spaces and, in particular, it can be used to calculate projections. As a result, $\mathcal{L}_2(v_1(u), v_2(u))$ gives a measure of how much $v_1(u)$ projects on $v_2(u)$, and the actual *geometric projection* length is obtained by scaling this value by the inverse of the norm of $v_2(u)$ (see again Eq. (1.10)). Note, however, that we need to ensure that the integral appearing in Eq. (1.11) is finite in order for the projection to be successful, but we do not insist on such aspect here.

The norm (or better the $\mathcal{L}_2$ -norm) of $v_2(u)$ itself can be obtained by taking the square root of the $\mathcal{L}_2$ inner product of $v_2(u)$ with itself, i.e. $\sqrt{\mathcal{L}_2(v_2(u), v_2(u))}$, exactly as the norm of a vector is given by the square root of the scalar product of the vector with itself, i.e. $|\mathbf{v}_2| = \sqrt{\mathbf{v}_2^T \mathbf{v}_2}$.

In summary, in both vector and function spaces, the scalar/inner product provides a quantitative measure of alignment between elements, which, upon appropriate normalisation, is equivalent to their components along a given direction. In a vector space, knowledge of the components of a vector along all independent directions amounts to the complete knowledge of the vector itself, whereas knowledge along only a few directions provides an incomplete characterisation. The same logic applies in function spaces, since functions can be regarded as infinite-dimensional vectors.

Assume now that we can generate a family of functions, $\mathcal{F}$, able to provide a basis of representation for an appropriate functional space, let us say polynomial functions. Such family is composed by monomials obtained as u^k with $k = 0, 1, 2, \dots, \infty$. Any polynomial can be expressed as a linear combination of these monomials, much like any vector can be expressed as a linear combination of its components.

We can obtain a characterisation of the PDF, $f(u)$, by calculating its projections on the elements of the basis $\mathcal{F}$. We denote these projections as *statistical moments*, m_k .

In particular, the statistical moment of order 0 (projection over the constant function), m_0 , is

$$m_0 = \int_{-\infty}^{\infty} f(u)u^0 du = \int_{-\infty}^{\infty} f(u)du = 1. \quad (1.13)$$

Thus, m_0 is the integral of $f(u)$ on the real axis, which represents the so-called normalisation condition. In particular, this integral must be 1, so expressing the certainty that a value will be extracted in the range $[-\infty, \infty]$.

Similarly, we can proceed to the moment of order 1, m_1 , leading to

$$m_1 = \int_{-\infty}^{\infty} f(u)u du = U, \quad (1.14)$$

showing that the first statistical moment is the *mean*, indicated by the capital letter U .

Proceeding in a similar way, we can obtain the second statistical moment

$$m_2 = \int_{-\infty}^{\infty} f(u)u^2 du = \text{RMS}^2(u), \quad (1.15)$$

which is indicated as the *mean square*. This value is usually expressed as the square of the *Root Mean Square*, RMS.

As a slight variation of the above procedure, it is common practice to define the so-called *central statistical moments*, shifting the probability distribution in order to centre it around its mean for moments higher than 1, leading to

$$m_{c2} = \int_{-\infty}^{\infty} f(u)(u - U)^2 du = \sigma_u^2, \quad (1.16)$$

where σ_u^2 is denoted as the *variance* of u . The square root of the variance is the *standard deviation* σ_u , which is a measure of dispersion around the mean and represents the typical fluctuation amplitude, see Fig. 1.12 (a).

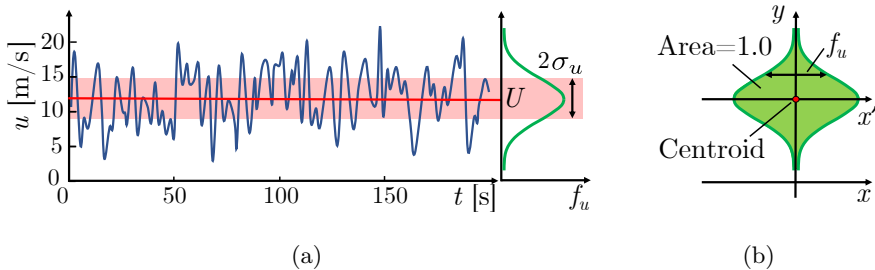


Figure 1.12: Statistical moments: (a) standard deviation as typical fluctuation and (b) beam section analogy.

As the strength of such fluctuations can be often more easily interpreted when compared to the mean itself, such information can be also expressed in terms of the *coefficient of variation*, σ_u/U .

It is useful to note the strong analogy existing between statistical moments and inertial properties of beam sections of unit area. In particular, assuming $f(u)$ to be the width of the section, m_0 is analogous to the section area, while m_1 and m_{c2} correspond to the static moment and inertia about the centroidal axis, respectively (see Fig. 1.12 (b)).

Proceeding in a similar way, we can define the *skewness*, Sk , as

$$m_{c3} = \int_{-\infty}^{\infty} f(u)(u - U)^3 du = Sk_u \sigma_u^3, \quad (1.17)$$

which is a measure of asymmetry of the PDF (see Fig. 1.13 (a)), in which the term σ_u^3 normalises Sk making it non-dimensional. Finally, we can define the *kurtosis*, Ku , as

$$m_{c4} = \int_{-\infty}^{\infty} f(u)(u - U)^4 du = Ku_u \sigma_u^4, \quad (1.18)$$

which is a measure of the tail thickness, i.e. how quickly the PDF decreases when moving away from the mean (see again Fig. 1.13 (b)). Note that, in order to maintain the same variance while increasing the tail thickness, the distribution must become “more peaked”.

All statistical moments have corresponding empirical estimators, i.e. a quantification of their values based on a finite number of observations, and

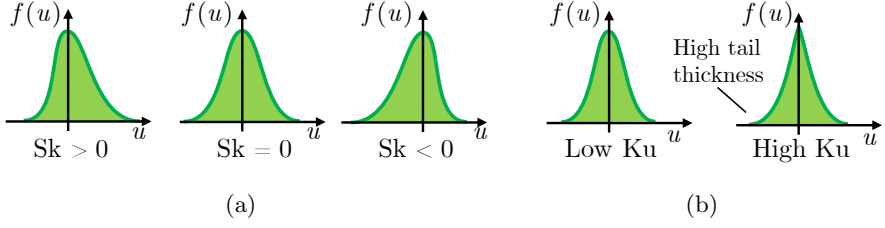


Figure 1.13: Shape of the PDF: (a) effects of skewness and (b) kurtosis.

they read

$$\hat{U} = \frac{1}{J} \sum_{j=1}^J \hat{u}_j, \quad \hat{\sigma}_u^2 = \frac{1}{J} \sum_{j=1}^J (\hat{u}_j - \hat{U})^2, \quad (1.19)$$

$$\widehat{Sk}_u = \frac{1}{\hat{\sigma}_u^3 J} \sum_{j=1}^J (\hat{u}_j - \hat{U})^3, \quad \widehat{Ku}_u = \frac{1}{\hat{\sigma}_u^4 J} \sum_{j=1}^J (\hat{u}_j - \hat{U})^4. \quad (1.20)$$

Note that these estimators closely resemble the integral definition provided above, but they are *biased*, i.e. they introduce systematic errors for small samples. In reality, *unbiased estimators* should be preferably used, for which the fraction appearing before the summation is modified incorporating the so-called Bessel's correction. For instance, for the variance, $J - 1$ should be used instead of J . However, if the sample size is sufficiently large, i.e. for large J , such corrected versions tend to those provided in Eq. (1.19) and Eq. (1.20).

In summary, statistical moments provide a synthetic description of the PDF, obtained from its projections onto monomial functions. These projections can be calculated theoretically from the PDF, but in practice they are estimated from a finite number of sample values, being the most important ones the mean and the standard deviation.

Additionally, it is often convenient to interpret the random variable u as the sum of its mean, U (often considered as deterministic), and its fluctuation around the mean u' (often considered as random). This leads to the splitting

$$u = U + u', \quad (1.21)$$

where u' is a random variable characterised by zero mean, i.e. $U' = 0$.

Note that, as the variance is by definition a centred statistical moment,

$\sigma_u^2 \equiv \sigma_{u'}^2$. This is not true for the RMS² and in particular

$$\text{RMS}^2(u) = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\hat{u}_j)^2 = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (U + \hat{u}'_j)^2 \quad (1.22)$$

$$= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [U^2 + 2U\hat{u}'_j + (\hat{u}'_j)^2] = U^2 + \sigma_{u'}^2, \quad (1.23)$$

$$= U^2 + \text{RMS}^2(u'), \quad (1.24)$$

where it has been used that $\frac{1}{J} \sum_{j=1}^J U\hat{u}'_j = \frac{U}{J} \sum_{j=1}^J \hat{u}'_j = UU' = 0$. Note that, going back to the analogy with the beam sections previously introduced, the result is equivalent to the Huygens-Steiner theorem.

1.4.3 Gaussian distribution

Fortunately, by knowing the nature of a random variable, it is often possible to assign *a priori* a particular statistical distribution. This happens, for instance, when we have information regarding its generating mechanism, or by knowing its relation to other random variables.

In particular, let us assume that our random variable is given by the sum of many independent random variables, none of which is dominant with respect to the others. Under fairly weak assumptions, as the number of summed variables increases, the distribution of their sum converges toward a *Gaussian distribution*. This fundamental result is known as the Central Limit Theorem, CLT.

The Gaussian distribution is characterised by CDF and PDF

$$F(u) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{u - U}{\sigma_u \sqrt{2}} \right) \right], \quad f(u) = \frac{1}{\sqrt{2\pi\sigma_u^2}} e^{-\frac{1}{2} \left(\frac{u - U}{\sigma_u} \right)^2}, \quad (1.25)$$

where $\text{erf}()$ is the *error function*. The equations above completely describe the distribution relying only on the value of the first two statistical moments, which is indeed another lucky fact.

Knowing the distribution also allows to define *confidence intervals*. Confidence intervals are of great practical importance, as they allow to define bounds between which we expect to find the value of the random variable at each extraction with known probability.

For instance, for the Gaussian, the interval defined by $2\sigma_u$ around the mean is characterised by a 95% probability to bound the random variable

at each extraction, because the integral under the curve in this interval is approximately equal to 0.95, see Fig. 1.14.

It can be shown that the Gaussian distribution is characterised by $S_k = 0$, like all symmetrical distributions, and $K_u = 3$. Sometimes the excess of Kurtosis with respect to the Gaussian value $K_{u_e} = K_u - 3$ is used instead of K_u to characterise the tail thickness, potentially causing misunderstandings.

As a first example of use of statistical moments, we can mention Gaussianity tests, such as the Jarque-Bera. If the sample is characterised by empirical $\widehat{S}_k$ and $\widehat{K}_u$ which remarkably deviate from the Gaussian values (i.e. 0 and 3, respectively), it is improbable that the underlying random variable follows a Gaussian distribution.

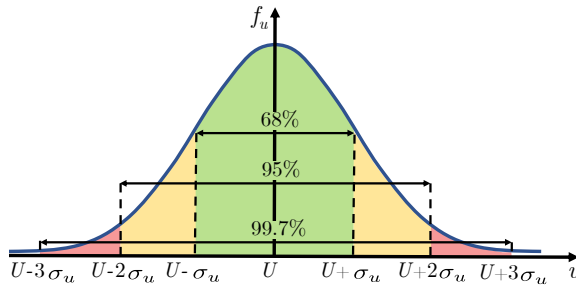


Figure 1.14: Gaussian distribution and corresponding confidence intervals.

In the case of wind, velocity fluctuations measured far from specific obstacles are influenced by a multitude of independent factors, consistent with the assumptions of the CLT. As a result, the fluctuations of the velocity vector components are expected to follow a Gaussian distribution.

It is thus possible to check if this is the case for a specific location using Gaussianity tests. If the test fails, the velocity components do not appear to follow a Gaussian distribution, probably indicating that a specific obstacle is strongly influencing the wind at the location under consideration, invalidating the assumptions underlying the CLT. Note that the procedure above applies to the velocity components, whereas the velocity magnitude is a non-linear combination of Gaussian-distributed components, so the CLT does not directly apply.

1.4.4 Sum of random variables

Let us now consider the case of a linear combination (i.e. weighted sum) of random variables, say $v = \beta_1 u_1 + \beta_2 u_2$. We wish to characterise its first

two statistical moments, V and σ_v , in terms of the statistical moments of u_1 and u_2 .

In order to calculate the mean of the weighted sum, we first evaluate its empirical counterpart, and then calculate the limit for $J \rightarrow \infty$ in order to obtain the true mean

$$V = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \hat{v}_j = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta_1 \hat{u}_{1,j} + \beta_2 \hat{u}_{2,j}) = \beta_1 U_1 + \beta_2 U_2, \quad (1.26)$$

which states that the mean of the weighted sum is the weighted sum of the means. We can now use Eq. (1.21) to isolate the fluctuating part of v , i.e. v' , and proceed to the calculation of its variance

$$\sigma_v^2 = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\hat{v}_j - V)^2 \quad (1.27)$$

$$= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [\beta_1 \hat{u}_{1,j} + \beta_2 \hat{u}_{2,j} - (\beta_1 U_1 + \beta_2 U_2)]^2 \quad (1.28)$$

$$= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta_1 \hat{u}'_{1,j} + \beta_2 \hat{u}'_{2,j})^2. \quad (1.29)$$

Expanding the square, we obtain

$$\sigma_v^2 = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [(\beta_1 \hat{u}'_{1,j})^2 + 2\beta_1 \beta_2 \hat{u}'_{1,j} \hat{u}'_{2,j} + (\beta_2 \hat{u}'_{2,j})^2] \quad (1.30)$$

$$= \beta_1^2 \sigma_1^2 + 2\beta_1 \beta_2 \text{cov}(u_1, u_2) + \beta_2^2 \sigma_2^2, \quad (1.31)$$

where $\text{cov}(u_1, u_2)$ is the *covariance* between u_1 and u_2 , defined as

$$\text{cov}(u_1, u_2) = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \hat{u}'_{1,j} \hat{u}'_{2,j}. \quad (1.32)$$

Looking at the equation above, it is clear that $\text{cov}(u_1, u_2) = \text{cov}(u_2, u_1)$, i.e. the covariance is symmetric. A graph showing the effect of covariance on two random variables is depicted in Fig. 1.15. If $\text{cov}(u_1, u_2) > 0$ the two variables tend to move together in the same direction, while the opposite is observed for $\text{cov}(u_1, u_2) < 0$. If $\text{cov}(u_1, u_2) \approx 0$ the variables are said to be *uncorrelated*, otherwise they are considered *correlated*.

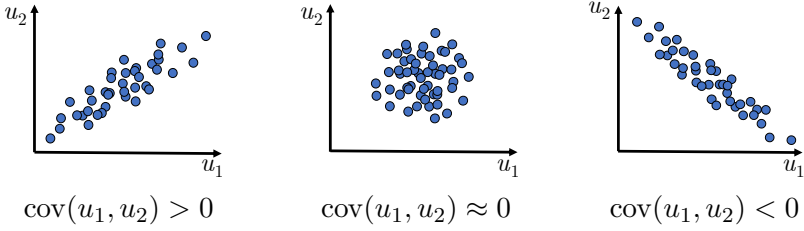


Figure 1.15: Correlated and uncorrelated random variables.

In summary, adopting matrix notation, we can rewrite Eq. (1.26) as

$$\mathbf{V} = \boldsymbol{\beta}^T \mathbf{U}, \quad (1.33)$$

where $\boldsymbol{\beta} = [\beta_1, \beta_2]$ and $\mathbf{U} = [U_1, U_2]$. Analogously, adopting an under-tilde for matrices, we can rewrite Eq. (1.31) as

$$\sigma_v^2 = \boldsymbol{\beta}^T \underline{\boldsymbol{\Sigma}}_u \boldsymbol{\beta}, \quad (1.34)$$

where, we have defined the *covariance matrix*, $\underline{\boldsymbol{\Sigma}}_u$, as

$$\underline{\boldsymbol{\Sigma}}_u = \begin{bmatrix} \sigma_{u,1}^2 & \text{cov}(u_1, u_2) \\ \text{cov}(u_1, u_2) & \sigma_{u,2}^2 \end{bmatrix}, \quad (1.35)$$

which is symmetric due to the symmetry of the covariance, see Eq. (1.32).

Adopting matrix notation, Eqs. (1.33) and (1.34) can be easily extended to evaluate the mean and variance of any linear combination of random variables. Note that independent variables, i.e. variables that do not share in any way their generating mechanisms, are always uncorrelated, but the opposite is not true. For instance, u and u^2 are uncorrelated if u is symmetrically-distributed. This is because the value of u^2 moves in the positive direction despite the sign of u . However, they are evidently not independent.

1.4.5 Correlation coefficient

It is of great interest to observe what happens when a random variable, v , is obtained by summation of a random variable of interest, u , and noise ϵ , assumed to be uncorrelated, i.e. $\text{cov}(u, \epsilon) = 0$. Here, by noise we mean a variable used to summarise everything we ignore about the generating

mechanism of v . In particular, we write

$$v = \beta u + \epsilon. \quad (1.36)$$

Applying Eq. (1.31), we obtain

$$\sigma_v^2 = \beta^2 \sigma_u^2 + \sigma_\epsilon^2, \quad (1.37)$$

We can then calculate the covariance between v and u

$$\text{cov}(v, u) = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \hat{v}'_j \hat{u}'_j = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta \hat{u}_j + \hat{\epsilon}_j)' \hat{u}'_j \quad (1.38)$$

$$= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [(\beta \hat{u}_j + \hat{\epsilon}_j) - (\beta U + \mathcal{E})]' \hat{u}'_j \quad (1.39)$$

$$= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta \hat{u}'_j + \hat{\epsilon}') \hat{u}'_j = \beta \sigma_u^2, \quad (1.40)$$

where we used the fact that $\text{cov}(u, \epsilon) = 0$. We can now define the *correlation coefficient*, ρ_{vu} , as

$$\rho_{vu} = \frac{\text{cov}(v, u)}{\sigma_v \sigma_u} = \frac{\beta \sigma_u^2}{\sigma_u \sqrt{\beta^2 \sigma_u^2 + \sigma_\epsilon^2}} \quad (1.41)$$

$$= \frac{\beta \sigma_u^2}{\sigma_u^2 \sqrt{\beta^2 + \sigma_\epsilon^2 / \sigma_u^2}} = \frac{\beta}{|\beta|} \frac{1}{\sqrt{1 + \sigma_\epsilon^2 / (\beta^2 \sigma_u^2)}}. \quad (1.42)$$

It can be thus seen that the absolute value of the correlation coefficient is equal to one if v is a simple scaling of u without the addition of noise (i.e. $\sigma_\epsilon^2 = 0$), and it decreases when noise prevails over the contribution of u . In the limit case in which only noise is present (i.e. $\beta = 0$), $\rho_{vu} = 0$ due to the infinite value attained by the term under square root at the denominator of Eq. (1.42).

Summarising, the correlation coefficient measures how much variance of v can be explained by a linear relation with u . By inspecting Eq. (1.42), we see that the correlation coefficient is bounded between -1 and 1, which indicate perfect negative and positive correlation, respectively.

The correlation coefficient can be also seen as a normalised covariance, as shown by its definition reported in Eq. (1.41). In particular, it represents the covariance normalised by the product of the standard deviations of the two variables. Readers wishing to deepen their knowledge of basic concepts in statistics beyond the requirements of the present discussion can refer to the specialised literature, for example [19].

1.5 Turbulence metrics

Let us now return to the velocity time histories measured at different heights above the ground, and concentrate on a single point in space, say P_1 , see again Fig. 1.6. We also assume the signal to be statistically stationary, meaning that its statistical descriptors do not evolve in time. We might use the PDF of the values recorded in the time history to provide a statistical characterisation. However, this is a lot of information, and it is usually unnecessary.

In fact, assuming the distribution to be Gaussian, it suffices to characterise the mean and the standard deviation to completely define the PDF. We now disregard the mean wind speed (already described in Sec. 1.3), which depends on the macro/meso-scale conditions, and focus exclusively on velocity fluctuations, which depend on micro-scale effects.

In particular, velocity fluctuations are usually characterised by means of their coefficient of variation

$$I_u = \frac{\sigma_u}{U}, \quad (1.43)$$

where I_u is the *turbulence intensity*. Turbulence intensity measures the strength of the velocity fluctuations relative to the mean, i.e. $I_u = 0.1 = 10\%$ indicates that typical velocity fluctuations are 10% of U . For example, looking at Fig. 1.14, it is possible to deduce that $[U - \sigma_u, U + \sigma_u] = [(1 - I_u)U, (1 + I_u)U]$ is a confidence interval which bounds u with 68% probability.

Let us now look at the two time histories reported in Fig. 1.16 (a). The two signals are clearly different, but they are characterised by a null mean value and the same standard deviation and, in fact, they oscillate approximately in the same range. The same considerations hold for the two signals reported in Fig. 1.16 (b). However, in this case *Signal B* seems to oscillate more rapidly than *Signal A*. This information is not contained in the PDF, so we need to introduce additional metrics in order to characterise such an aspect. Furthermore, the fact that the signal is non-periodic prevents us from using the simple notion of frequency.

With the intention of deriving a quantitative descriptor for the fluctuation speed, we select an instant in time, say t , and a time lag, say τ . The signal can then be decomposed in the form

$$u(t + \tau) = \beta u(t) + \epsilon, \quad (1.44)$$

where β is a constant. The value of u at time $t + \tau$ is thus seen as the sum of a term proportional to the old value at time t and an uncorrelated term

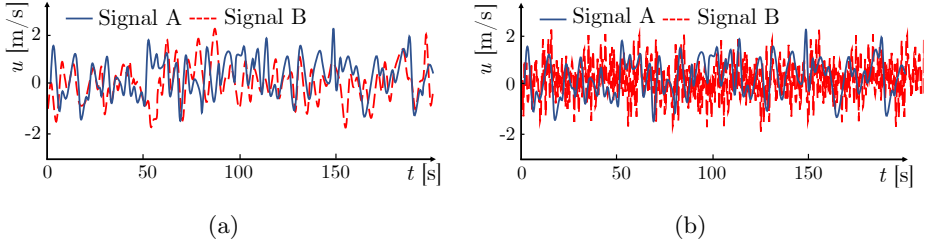


Figure 1.16: Examples of signals having: (a) the same integral time-scale and (b) different integral time-scale.

ϵ , which is also known as *innovation*.

We can measure how much of the variance of $u(t+\tau)$ depends on $u(t)$ by means of the correlation coefficient $\rho_{u(t),u(t+\tau)}$. We thus proceed to calculate the correlation coefficient between the signal and itself at different time lags. The function describing $\rho_{u(t),u(t+\tau)}(\tau) = \rho_{uu}(\tau)$ is the *autocorrelation function*, ACF. For $\tau = 0$ no innovation can be present, and so $\rho_{uu}(0) = 1$.

In the case of wind and other phenomena which do not show dominant frequencies (i.e. they are broad-banded), the ACF can be well approximated by an exponential decay

$$\rho_{uu}(\tau) = e^{-\tau/T}, \quad (1.45)$$

where T is a parameter regulating the speed of the decay, see Fig. 1.17 (b).

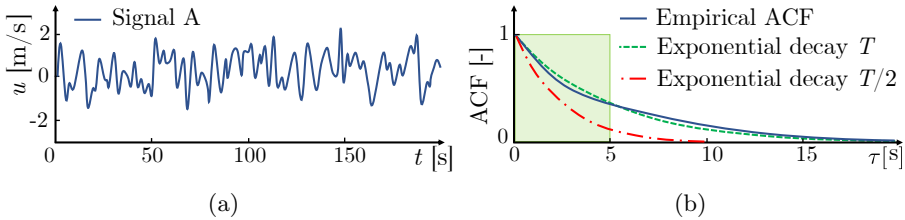


Figure 1.17: Calculation of the autocorrelation function: (a) signal and (b) ACF.

It is of interest to condense in a single number the most important information provided by $\rho_{uu}(\tau)$. In particular, we can define the *integral time-scale* of the velocity component u , T_u , as the area under the ACF

$$T_u = \int_0^\infty \rho_{uu}(\tau) d\tau. \quad (1.46)$$

A useful graphical interpretation is to consider T_u as the base of the rectangle which has unit height and area equal to that underlying the ACF, see the rectangle shaded in Fig. 1.17 (b), which assumes $T_u = 5$. From this point of view, T_u represents a characteristic time-lag. In particular, fluctuations separated in time by less than T_u can be considered as fully correlated, while those separated by longer lags can be considered fully uncorrelated.

Note that the possibility to use T_u as a meaningful synthetic descriptor of the ACF is possible mainly because we have an *a priori* expectation regarding its shape, i.e. the exponential decay. The line of reasoning is, in fact, similar to the one already followed for the characterisation of the PDF of the wind fluctuations. In that case, basing the description only on U and I_u relies on the *a priori* expectation of Gaussianity.

In particular, in the case of an ACF described by an exponential decay, we have

$$T_u = \int_0^\infty \rho_{uu}(\tau) d\tau = \int_0^\infty e^{-\tau/T} d\tau = \left[-T e^{-\tau/T} \right]_0^\infty = T, \quad (1.47)$$

showing that the T appearing in the exponential decay in Eq. (1.45) actually coincides with the integral time-scale. Looking back to Fig. 1.16 (b), it can be now noted that *Signal A* has a longer integral time-scale with respect to *Signal B*.

Furthermore, in the provided example, the ACF is calculated by considering a time lag τ while measuring the signal at the same location. However, the concept can be immediately extended to synchronous measurements at points separated in space.

For example, let us consider Fig. 1.18, which might represent the velocity field in the ABL. As it can be seen, the flow has a random pattern, composed of bigger and smaller structures. In cases of our interest, these structures are usually vortices, superimposed one another. This is in fact our first encounter with a turbulent flow, although greatly simplified in the drawing. The flow can be imagined as a superposition of eddies transported by the time-averaged velocity field. This idealisation is called Taylor's *frozen turbulence* assumption, and we will encounter it again in due time.

To extend the concept of integral scale from time to space, we consider two points, say P_1 and P_2 separated along the x - direction by distance ξ .

Chapter 2

Fluid flow governing equations

...In the previous chapter, we set the stage for our investigation. It is now time to become familiar with the fundamental rules that govern our environment and deepen our understanding of turbulence, which we have already encountered briefly. The same phenomenon can often be observed at different scales, yielding seemingly unrelated pictures. Order is established only by relating these pictures to one another...

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2.1 Balance equations

We begin by considering a volume of fluid, V , bounded by a surface S , as shown in Fig. 2.1 (a). The volume is fixed in space, while fluid flows around and within it, eventually crossing S .

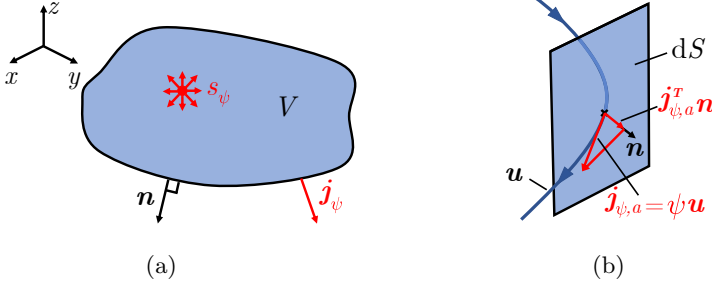


Figure 2.1: Balance equations: (a) overview of the control volume and (b) advective flux through an elementary portion of the control surface.

Let us assume some scalar quantity is defined over the volume, say ψ , which might be an ink concentration if you wish. The scalar field ψ represents a density per unit volume. As the volume is used in order to check how much ψ enters and exits from it, we denote V as the *control volume* and S as the *control surface*.

We first need to measure how much ψ we have inside the control volume. We can calculate it as

$$\int_V \psi dV. \quad (2.1)$$

This amount might change in time due to different phenomena. In particular, some ψ might flow through the volume surfaces and some might be created or destroyed within V . We indicate the *flux* of ψ , i.e. the amount of ψ transferred through a unit portion of the surface S per unit time, as $\mathbf{j}_\psi$. We can calculate the total amount of ψ exiting the control volume per unit time as

$$\int_S \mathbf{j}_\psi^T \mathbf{n} dS, \quad (2.2)$$

where $\mathbf{n}$ is the outward pointing normal. The scalar product $\mathbf{j}_\psi^T \mathbf{n}$ gives the component of $\mathbf{j}_\psi$ normal to the surface, which is the only part that actually crosses S , and thus leaves V . Considering that the normal has been taken as oriented outward, the term represents an outward-oriented flux, i.e. a loss of ψ from the control volume.

Additionally, we need to account for the amount of ψ which is created within the control volume. This can be calculated as

$$\int_V s_\psi dV, \quad (2.3)$$

where s_ψ is the *production*, or *destruction* if negative, of ψ per unit volume per unit time.

The time variation of the contribution expressed in Eq. (2.1) must be equal to the sum of those in Eq. (2.2) and (2.3), leading to

$$\frac{d}{dt} \int_V \psi dV = - \int_S \mathbf{j}_\psi^T \mathbf{n} dS + \int_V s_\psi dV, \quad (2.4)$$

where we stress again that the contribution arising from surface fluxes is negative because it represents a loss. Equation (2.4) is equally valid for control volumes of fluid and solid nature. However, for fluids, it is convenient to split the flux $\mathbf{j}_\psi$ into two contributions

$$\mathbf{j}_\psi = \mathbf{j}_{\psi,a} + \mathbf{j}_{\psi,na}, \quad (2.5)$$

where $\mathbf{j}_{\psi,a}$ is the advective part of $\mathbf{j}_\psi$, while $\mathbf{j}_{\psi,na}$ takes into account all other possible flux contributions. In particular, the advective part accounts for the amount of ψ leaving V transported by the fluid flowing through S , which can be written

$$\mathbf{j}_{\psi,a} = \psi \mathbf{u}, \quad (2.6)$$

where $\mathbf{u} = [u, v, w]^T$ is, as usual, the velocity vector, see Fig. 2.1 (b).

Conversely, non-advective fluxes represent all other transport mechanisms of ψ , most often the natural tendency of a quantity to spread from regions of high concentration toward regions of lower concentration even in absence of fluid motion, i.e. *diffusion*.

It is finally common to move the flux terms to the left of the equation leading to

$$\frac{d}{dt} \int_V \psi dV + \int_S \psi \mathbf{u}^T \mathbf{n} dS + \int_S \mathbf{j}_{\psi,na}^T \mathbf{n} dS = \int_V s_{\psi,p} - s_{\psi,d} dV, \quad (2.7)$$

in which the *source term* s_ψ has been expressed as $s_\psi = s_{\psi,p} - s_{\psi,d}$ where $s_{\psi,p}$ is the production of ψ (i.e. a positive source term), while $s_{\psi,d}$ represents its destruction (i.e. a sink term).

Equation (2.7) provides a clear understanding of all the main phenomena which take place in fluid flows: time variation is given by advection, other fluxes (e.g. diffusive ones), production and destruction. This equation represents the *balance* of ψ in integral form.

If $\mathbf{j}_\psi$ and $\mathbf{u}$ are sufficiently smooth, which is always the case for our applications, we can transform surface integrals into volume integrals using the Gauss-Green theorem (further details given in Sec. 5.1.3). In particular, the theorem ensures that

$$\int_S \gamma^T \mathbf{n} dS = \int_V \nabla \cdot \gamma dV, \quad (2.8)$$

where $\cdot$ is the dot product, γ here represents a generic vector field defined over the control volume and *nabla*, $\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]^T$, is a differential operator. Essentially, as the dot product symbol, $\cdot$, is present, first we transpose the ∇ operator making it a row, then we proceed to the usual matrix multiplication. More in general, considering ∇ as a usual vector, under our convention, the dot product coincides with the scalar product when acting between two vectors, while it behaves as a scalar-vector multiplication in other circumstances. The reader, which might have been previously exposed to other conventions, is therefore advised to keep this non-standard usage in mind to avoid confusion. We thus have

$$\nabla \cdot \gamma = \begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{bmatrix} \begin{bmatrix} \gamma_x \\ \gamma_y \\ \gamma_z \end{bmatrix} = \frac{\partial \gamma_x}{\partial x} + \frac{\partial \gamma_y}{\partial y} + \frac{\partial \gamma_z}{\partial z}, \quad (2.9)$$

a quantity known as the *divergence* of γ . Using the Gauss-Green theorem taking $\gamma = \psi \mathbf{u}$, we can rewrite Eq. (2.7) as

$$\frac{d}{dt} \int_V \psi dV + \int_V \nabla \cdot (\psi \mathbf{u}) dV + \int_V \nabla \cdot \mathbf{j}_{\psi,na} dV = \int_V s_{\psi,p} - s_{\psi,d} dV. \quad (2.10)$$

In the equation above, the time derivative can enter the integral and the terms can be grouped together, leading to

$$\int_V \left[\frac{\partial \psi}{\partial t} + \nabla \cdot (\psi \mathbf{u}) + \nabla \cdot \mathbf{j}_{\psi,na} \right] dV = \int_V [s_{\psi,p} - s_{\psi,d}] dV. \quad (2.11)$$

Note that, when entering the integral, the total time derivative becomes a partial derivative, as spatial dimensions are no longer eliminated by the integration.

The equation above must be true for all possible control volumes, so that the integrand on the left and right sides must balance, leading to

$$\frac{\partial \psi}{\partial t} + \nabla \cdot (\psi \mathbf{u}) + \nabla \cdot \mathbf{j}_{\psi,na} = s_{\psi,p} - s_{\psi,d}, \quad (2.12)$$

which is the balance equation for ψ in differential form [22, 23].

2.1.1 Mass balance

We can now apply the balance equation reported in Eq. (2.12) to obtain the equations governing the motion of fluids. In particular, let us apply the balance equation to mass, obtaining the mass balance.

In this case, we set $\psi = \rho$ with ρ fluid density (i.e. mass per unit volume) and obtain

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) + \nabla \cdot \mathbf{j}_{\rho,na} = s_{\rho,p} - s_{\rho,d}. \quad (2.13)$$

For our purposes, we can assume that mass is conserved, so that $s_{\rho,p} = s_{\rho,d} = 0$. Note that this might not be the case if ρ represents the density of a particular chemical species and chemical reactions occur. Additionally, we introduce the Mach number, Ma , representing the ratio between the fluid speed and the speed of sound in the fluid. For low Ma , say $Ma < 0.3$, we can neglect compressibility, so that ρ is constant in both space and time. It follows that, in the low Ma range, liquids and gases behave in the same way, i.e. they are *incompressible*. Considering that the speed of sound in air is 343 m/s while wind velocity is typically below 60 m/s, the approximation is well acceptable for our cases.

Furthermore, for the mass balance equation, $\mathbf{j}_{\rho,na}$ is a null vector (i.e. the mass flux is only given by advection if the fluid is composed of a single species or it is a homogeneous mixture). In these conditions, Eq. (2.13) becomes

$$\nabla \cdot \mathbf{u} = 0, \quad (2.14)$$

or, in components

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad (2.15)$$

showing that mass conservation for incompressible fluids leads to a constraint of null divergence for the velocity field, i.e. the flow is *divergence-free*.

It is now useful to obtain a more intuitive understanding of the origin of

the mass balance equation. In particular, let us consider the parallelepiped-shaped elementary volume depicted in Fig. 2.2.

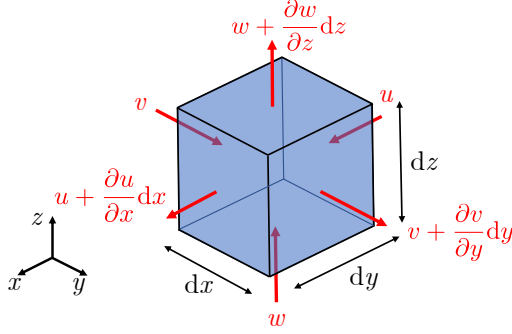


Figure 2.2: Elementary parallelepiped control volume reporting the normal velocities over its faces.

Again the fluid is assumed to be incompressible, so that the volume of fluid entering and exiting across the surfaces per unit time must sum to zero. For instance, the volume exiting from the face with normal oriented along the positive x -direction is $(u + \frac{\partial u}{\partial x} dx) dy dz$, while a volume $u dx dy dz$ enters from the face oriented along the negative x -direction. Summing over all faces, the balance can be written as

$$(u + \frac{\partial u}{\partial x} dx - u) dy dz + (v + \frac{\partial v}{\partial y} dy - v) dx dz + (w + \frac{\partial w}{\partial z} dz - w) dx dy = 0, \quad (2.16)$$

which, simplifying terms and dividing by $dx dy dz$ leads to

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad (2.17)$$

which coincides with Eq. (2.14).

2.1.2 Momentum balance

We can also impose the balance of *momentum* (i.e. quantity of motion). Unlike density, which is a scalar quantity, the momentum per unit volume is a vector defined as $\rho \mathbf{u}$, for which we can write a balance equation for each component. For instance, replacing ψ in Eq. (2.12) with the first component of such a vector, ρu , yields

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{u}) + \nabla \cdot \mathbf{j}_{\rho u, na} = s_{\rho u, p} - s_{\rho u, d}. \quad (2.18)$$

In this case we have $s_{\rho u, p} = f_x$ where f_x is the force per unit volume in the x -direction arising, for instance, from gravity or Coriolis force, while $s_{\rho u, d} = 0$. Additionally, the non-advective fluxes of momentum through the control surface, $\mathbf{j}_{\rho u, na}$, are represented by the *tractions*, $\mathbf{t}$, applied there. Tractions can be obtained from the stress tensor, $\underline{\sigma}$, as

$$\mathbf{t} = \underline{\sigma} \mathbf{n}, \quad (2.19)$$

where it can be shown that $\underline{\sigma}$ is symmetric by considering the conservation of angular momentum. The argument closely follows the one used in solid mechanics to demonstrate that $\underline{\sigma}$ is symmetric as a consequence of rotational equilibrium for Cauchy continua. We thus have

$$\underline{\sigma} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix}. \quad (2.20)$$

In particular, as we are considering the balance of momentum in the x -direction only, we need to extract the first component of the traction, t_x . We can obtain it as

$$t_x = \mathbf{e}_x^T \mathbf{t} = \mathbf{e}_x^T (\underline{\sigma} \mathbf{n}), \quad (2.21)$$

where $\mathbf{e}_x = [1, 0, 0]^T$ is the unit vector oriented in the x -direction. We can, however, group terms differently obtaining

$$t_x = \mathbf{e}_x^T (\underline{\sigma} \mathbf{n}) = (\mathbf{e}_x^T \underline{\sigma}) \mathbf{n} = \underline{\sigma}[1, :] \mathbf{n}, \quad (2.22)$$

where $\underline{\sigma}[1, :]$ is the first row of $\underline{\sigma}$. As a result, we have that the non-advective flux of x -directed momentum can be calculated as

$$t_x = \underline{\sigma}[1, :] \mathbf{n} = -\mathbf{j}_{\rho u, na}^T \mathbf{n}, \quad (2.23)$$

where the minus sign comes from the fact that, according to the sign convention adopted for stresses, positive contributions act as inward fluxes. Using the symmetry of the stress tensor, the equation above allows us to identify

$$\mathbf{j}_{\rho u, na} = -(\underline{\sigma}[1, :])^T = -\underline{\sigma}[:, 1] = -[\sigma_x, \tau_{xy}, \tau_{xz}]^T. \quad (2.24)$$

The fact that momentum fluxes are statically equivalent to negative stresses is further illustrated in Fig. 2.3. Specifically, the reaction engine of the rocket uses combustion to generate an outward-directed momentum flux, which accelerates the rocket in the opposite direction, as shown in Fig. 2.3 (a). This effect is *statically equivalent* to applying inward-directed normal stresses, as depicted in Fig. 2.3 (b).

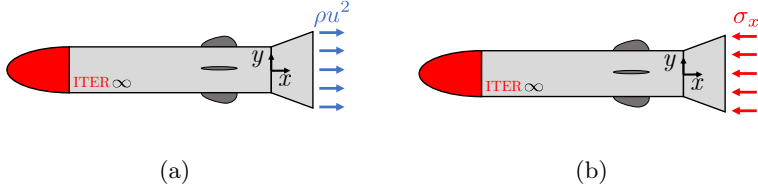


Figure 2.3: Static equivalence between momentum fluxes and stresses: (a) outward-directed momentum flux and (b) inward-directed normal stresses.

The x -directed momentum balance equation can be thus rewritten as

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{u}) - \nabla \cdot (\boldsymbol{\sigma}[:, 1]) = f_x. \quad (2.25)$$

Still assuming the flow density to be constant, i.e. incompressibility, we obtain

$$\frac{\partial u}{\partial t} + \nabla \cdot (u \mathbf{u}) - \frac{1}{\rho} \nabla \cdot (\boldsymbol{\sigma}[:, 1]) = \frac{f_x}{\rho}. \quad (2.26)$$

We can further observe that, following the usual rules for the derivative of the product between two functions

$$\nabla \cdot (u \mathbf{u}) = (\nabla \cdot u) \mathbf{u} + u (\nabla \cdot \mathbf{u}) = (\nabla \cdot u) \mathbf{u}, \quad (2.27)$$

where $u \nabla \cdot \mathbf{u} = 0$ accounting for Eq. (2.14). We further notice that, according to our convention

$$\nabla \cdot u = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right] u = \left[\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right], \quad (2.28)$$

showing that $\nabla \cdot u$ is the gradient of u expressed as a row vector, which we denote as $\nabla(u)$. We thus arrive at the form

$$\frac{\partial u}{\partial t} + \nabla(u) \mathbf{u} - \frac{1}{\rho} \nabla \cdot \boldsymbol{\sigma}[:, 1] = \frac{f_x}{\rho}, \quad (2.29)$$

in which $\frac{1}{\rho} \nabla \cdot \boldsymbol{\sigma}[:, 1]$ is the divergence of the stresses acting in the x -direction and, in particular

$$\nabla \cdot \boldsymbol{\sigma}[:, 1] = \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z}. \quad (2.30)$$

Up to now, we derived the momentum balance equation only for the x -direction. However, looking at its structure, it is easy to infer that, for

the y -direction it reads

$$\frac{\partial v}{\partial t} + \nabla(v) \mathbf{u} - \frac{1}{\rho} \nabla \cdot \boldsymbol{\varrho}[:, 2] = \frac{f_y}{\rho}, \quad (2.31)$$

and similarly for the z -direction.

The momentum balance equations can be, thus, collectively represented by means of a vector equation

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u}) \mathbf{u} - \frac{1}{\rho} \nabla \cdot \boldsymbol{\varrho} = \frac{\mathbf{f}}{\rho}. \quad (2.32)$$

Note that $\nabla(\mathbf{u})$ in the previous equation is a matrix, even though the tilde to denote this is omitted, tacitly understanding that the gradient is calculated for each component of $\mathbf{u}$. Additionally, the careful reader might note that passing from Eq. (2.29) and Eq. (2.31), i.e. from component by component to the vectorial notation in Eq. (2.32), would require some adaptation. In particular, according to our notation, we would need to write $(\nabla \cdot \boldsymbol{\varrho})^T$ instead of the simple $\nabla \cdot \boldsymbol{\varrho}$, i.e. we desire in output a column vector. However, omitting the transposition is common practice in the literature and the dot product of ∇ with a second-order tensor is understood to yield its divergence as a column vector.

In particular, expressing Eq. (2.32) in components, we have

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \\ w \end{pmatrix} + \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} - \frac{1}{\rho} \begin{bmatrix} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \end{bmatrix} = \frac{1}{\rho} \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}, \quad (2.33)$$

or, separating them

$$\begin{cases} \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right] = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} u + \frac{\partial v}{\partial y} v + \frac{\partial v}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right] = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x} u + \frac{\partial w}{\partial y} v + \frac{\partial w}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \right] = \frac{f_z}{\rho}. \end{cases} \quad (2.34)$$

It is now useful to obtain a more intuitive understanding of this result. Let us consider again only the balance of momentum in the x -direction and the elementary volume reported in Fig. (2.4). We observe that there is an abuse of notation in the figure. In fact the same value, u , has been assigned to all faces with a negative normal, but they should be actually distinguished because they are measured at different locations. In a similar way, we should differentiate also the symbol used for the value at the parallelepiped centre.

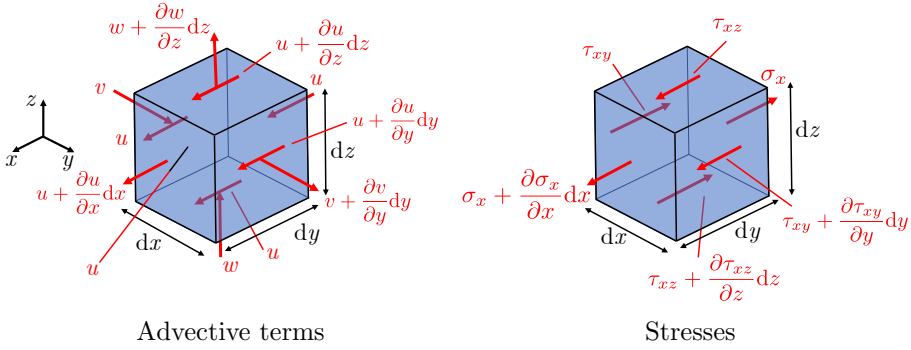


Figure 2.4: Elementary volume of fluid, normal velocities and stresses over its faces.

This would, however, lead to numerous terms and, fortunately, results are not affected by the adopted simplification which is, therefore, retained.

We again assume incompressibility and we begin by considering the rate of change of momentum within the control volume. This is

$$\frac{\partial(\rho u)}{\partial t} dx dy dz = \rho \frac{\partial u}{\partial t} dx dy dz. \quad (2.35)$$

We now calculate the net flux of momentum due to advection through the faces whose normal is directed along the x -direction. We have

$$\left[\left(u + \frac{\partial u}{\partial x} dx \right)^2 - u^2 \right] \rho dy dz. \quad (2.36)$$

We can expand the square, simplify terms and disregard higher-order contributions (i.e. those of fourth-order in the parallelepiped dimensions), so that Eq. (2.36) becomes

$$2u \frac{\partial u}{\partial x} \rho dx dy dz. \quad (2.37)$$

In a similar way, we can proceed to calculate the net momentum flux due to advection through the faces which have normal along y . In this case, we write

$$\left[\left(u + \frac{\partial u}{\partial y} dy \right) \left(v + \frac{\partial v}{\partial y} dy \right) - uv \right] \rho dx dz. \quad (2.38)$$

The contribution can be rewritten neglecting higher-order terms, leading to

$$\left(u \frac{\partial v}{\partial y} + v \frac{\partial u}{\partial y} \right) \rho dx dy dz. \quad (2.39)$$

It is easy to imagine that the net momentum flux due to advection through the faces which have normal along z have a similar form to that found in Eq. (2.39). Summing all the advective contributions, we obtain

$$\begin{aligned} \left[2u \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} u + \frac{\partial u}{\partial y} v + \frac{\partial w}{\partial z} u + \frac{\partial u}{\partial z} w \right] \rho dx dy dz = \\ = \left[\frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w \right] \rho dx dy dz, \end{aligned} \quad (2.40)$$

where, in the last passage, the mass conservation equation has been used to delete two terms and eliminate the factor 2 on the first one.

Finally, we consider the contribution due to stresses and volume forces. In particular, summing the forces exchanged by all faces, we have

$$\begin{aligned} \left[\sigma_x + \frac{\partial \sigma_x}{\partial x} dx - \sigma_x \right] dy dz + \left[\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y} dy - \tau_{xy} \right] dx dz \\ + \left[\tau_{xz} + \frac{\partial \tau_{xz}}{\partial z} dz - \tau_{xz} \right] dx dy, \end{aligned} \quad (2.41)$$

which can be simplified as

$$\left[\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right] dx dy dz. \quad (2.42)$$

Finally, we can calculate the contribution due to volume forces as

$$f_x dx dy dz. \quad (2.43)$$

Adding with the appropriate signs the terms reported in Eq. (2.35), Eq. (2.40), Eq. (2.42), Eq. (2.43), and dividing by $\rho dx dy dz$ leads again to the first of Eq. (2.34).

It is finally useful to note, without proof, that in the case of incompressible fluids and when heat transport is not considered, the energy balance does not provide additional information beyond that already contained in the mass and momentum balances. The energy equation is in fact a linear combination of the others.

2.1.3 Total derivative

We have yet another approach to visualise the momentum balance equation. Up to now we considered the control volume to be fixed in space, i.e. adopting an Eulerian approach.

We now adopt a Lagrangian approach and assume the control volume to move with the flow. From this perspective, the control volume can be considered like a particle transported by the flow, so we denote it as fluid particle, assumed of mass m . We can apply Newton's second law of motion writing

$$m\mathbf{a} = \mathbf{r}, \quad (2.44)$$

where $\mathbf{a}$ is the fluid particle acceleration and $\mathbf{r}$ is the resultant of all volume and surface forces applied to the fluid particle. Note that $\mathbf{r}$ has the dimensions of a force, whereas $\mathbf{f}$ previously introduced is a force density. We can rewrite Eq. (2.44) as

$$m \frac{D\mathbf{u}}{Dt} = \mathbf{r}, \quad (2.45)$$

where $\frac{D}{Dt}$ indicates the *total derivative*, i.e. the actual variation in velocity “felt” by the fluid particle during its motion. In particular, the total derivative of the first component of $\mathbf{u}$ is

$$\frac{Du}{Dt} = \frac{u(\mathbf{x}(t+dt), t+dt) - u(\mathbf{x}(t), t)}{dt}, \quad (2.46)$$

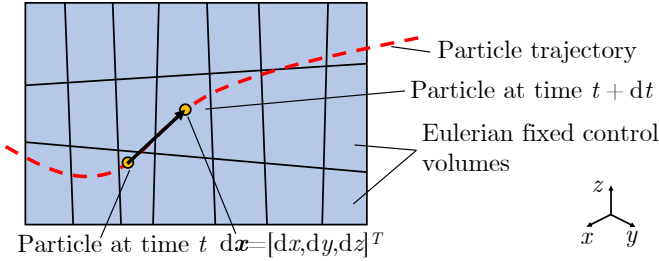


Figure 2.5: A fluid particle moving along its trajectory: Eulerian and Lagrangian perspectives.

We recall that the Eulerian velocity is measured at fixed locations in space and can be interpreted as a background velocity field describing, at every point, the motion of the fluid particles passing beneath the measurement location, see Fig. 2.5. Thus, the fluid particle position at time $t+dt$ is

$$\mathbf{x}(t+dt) = \mathbf{x}(t) + \mathbf{u}dt. \quad (2.47)$$

We can therefore rewrite Eq. (2.46) as

$$\frac{Du}{Dt} = \frac{u(x + udt, y + vdt, z + wdt, t+dt) - u(x, y, z, t)}{dt}, \quad (2.48)$$

which can be written via a first-order Taylor expansion as

$$\frac{Du}{Dt} = \frac{u(x, y, z, t) + \frac{\partial u}{\partial x} u dt + \frac{\partial u}{\partial y} v dt + \frac{\partial u}{\partial z} w dt + \frac{\partial u}{\partial t} dt - u(x, y, z, t)}{dt}, \quad (2.49)$$

leading to

$$\frac{Du}{Dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w = \frac{\partial u}{\partial t} + \nabla(u) \cdot \mathbf{u}. \quad (2.50)$$

In summary, while moving on its trajectory, the velocity of the fluid particle varies due to the evolution in time of the Eulerian velocity field, as well as variations due to the particle advancement in space. If the fluid particle could stay still in space, it would experience only the first contribution. Conversely, if the particle advances from low to high velocity zones, even if the background Eulerian velocity field does not change in time, the particle experiences this as an acceleration.

The evaluation of the other terms occurring in Eq. (2.44) does not deviate substantially from that previously presented, being $\mathbf{r}$ as anticipated the combined effect of surfaces stresses and volume forces. Summing all contributions, the momentum conservation equation is obtained again, as already presented in Eq. (2.33).

Comparing Eq. (2.50) to the first terms of Eq. (2.33), we see that the time derivative and the advective terms collectively represent inertial terms, i.e. the total derivative of the momentum of the fluid particle, moving along the path defined by the Eulerian velocities.

Finally, it is useful to note that the total derivative can be calculated for any quantity, say ψ , and reads

$$\frac{D\psi}{Dt} = \frac{\partial \psi}{\partial t} + \frac{\partial \psi}{\partial x} u + \frac{\partial \psi}{\partial y} v + \frac{\partial \psi}{\partial z} w = \frac{\partial \psi}{\partial t} + \nabla(\psi) \cdot \mathbf{u}. \quad (2.51)$$

2.2 Euler equations

Up to now, we have not detailed how the stress tensor $\underline{\sigma}$ should be calculated. It is well known that a fluid at rest develops only a stress state such that

$$\underline{\sigma} = -p\mathbf{I}, \quad (2.52)$$

where p is pressure and $\mathbf{I}$ is the identity matrix (of size 3). The minus stems from the fact that pressure is assumed to be positive when compressing the control volume. With this assumption, it can be easily verified that

$$\nabla \cdot \underline{\sigma} = -\nabla p. \quad (2.53)$$

Note that ∇p , following the usual rule of vector times scalar multiplication, is the gradient of p expressed as a column vector.

If the fluid does not move, i.e. $\mathbf{u} = \mathbf{0}$ everywhere at every time, Eq. (2.32) becomes

$$\nabla p = \mathbf{f}, \quad (2.54)$$

which is, indeed, the equation describing fluids at rest.

Furthermore, even when the fluid is in motion, the pressure gradient often dominates the stress field. Under this assumption, momentum conservation reads

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u}) \mathbf{u} + \frac{1}{\rho} \nabla p = \frac{\mathbf{f}}{\rho}, \quad (2.55)$$

or, in components, starting from Eq. (2.34) and adding as first equation the mass conservation expressed in Eq. (2.15)

$$\begin{cases} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \\ \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} u + \frac{\partial v}{\partial y} v + \frac{\partial v}{\partial z} w + \frac{1}{\rho} \frac{\partial p}{\partial y} = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x} u + \frac{\partial w}{\partial y} v + \frac{\partial w}{\partial z} w + \frac{1}{\rho} \frac{\partial p}{\partial z} = \frac{f_z}{\rho}. \end{cases} \quad (2.56)$$

This set of equations is known as Euler equations. They consist of four equations in four unknowns, making the system, in principle, solvable. Three of the equations are *evolution equations* (i.e. they contain the time derivative), obtained from momentum balance, while mass conservation acts as a *constraint*, enforcing incompressibility.

It is worth noting, without proof, that for incompressible flows, the pressure field can be interpreted as a Lagrange multiplier enforcing the mass conservation constraint on the momentum equations, i.e. pressure always adjusts so that momentum equations yield a divergence-free velocity field.

2.3 Molecular viscosity

Euler equations neglect an important contribution to the non-advective momentum flux: *viscous stresses*. While these stresses might be dominated by pressure in many circumstances, their contribution is fundamental to explain the dynamics of fluids.

At the macroscopic level, viscous stresses might be described as a sort of friction between adjacent fluid layers, see Fig. 2.6 (a) and (b). In particular, the fluid is moving in the x -direction. However, two fluid layers adjacent to each other are separated by a distance dy and, thus, are characterised by

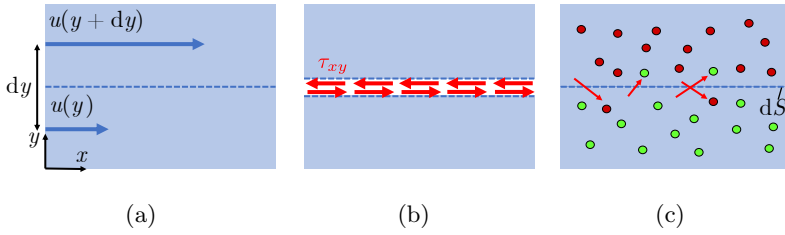


Figure 2.6: Effect of viscosity seen at the macroscopic and microscopic scales: (a) adjacent fluid layers, (b) exchanged shear stress and (c) molecular random motion.

slightly different velocities, i.e. $u(y)$ and $u(y+dy)$. Newton noticed that viscous stresses can be evaluated as

$$\tau_{xy} = \mu \frac{du}{dy}, \quad \text{or alternatively} \quad \tau_{xy} = \nu \rho \frac{du}{dy}, \quad (2.57)$$

where μ is the *viscosity coefficient*, while $\nu = \mu/\rho$ is the *kinematic viscosity coefficient*. If such formulation is applicable with good approximation, the fluid is said to be Newtonian. Water and air are good examples of Newtonian fluids.

We now seek a micromechanical explanation of this result. Looking at Fig. 2.6 (c), we can see that, at the molecular level, some particles migrate from one fluid layer to the adjacent one due to random thermal motions [24].

This migration of molecules leads to a transfer of momentum, which accelerates the low-velocity fluid layer and decelerates the high-velocity one. At the macroscopic scale this manifests as a momentum flux and, thus, a shear stress which can be seen to be proportional to the derivative of the velocity. In particular, to quantify this momentum flux, we assume that

- i) molecules are characterised by the velocity typical of the fluid layer they come from;
- ii) they typically move at the *thermal speed*, $|\mathbf{u}_{th}|$, relative to the fluid layer overall motion;
- iii) the direction of the thermal motion is random and equally probable in all directions;
- iv) typically, a molecule travels a distance λ , the *mean free path*, before colliding with another particle, so equalising its speed with the surrounding fluid molecules.

With reference to Fig. 2.7 (a), we notice that at any point half molecules move upwards and half move downwards, due to the fact that we assumed all directions to be equally probable for the thermal motion. Furthermore, the average vertical velocity of the particles moving upward can be easily shown to be $|\mathbf{u}_{th}|/2$ (see again Fig. 2.7 (a) where the value is the average vertical length of the upward directed arrows).

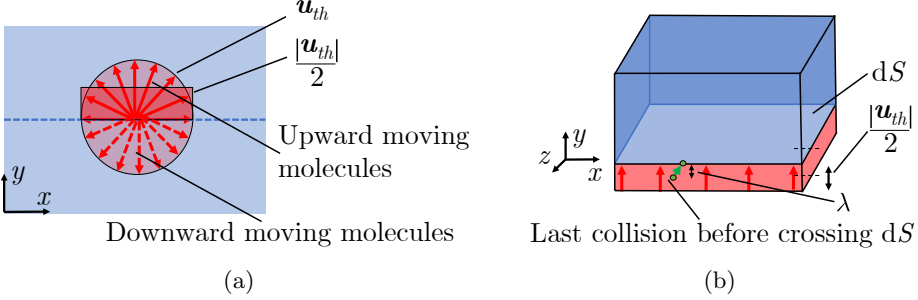


Figure 2.7: Micromechanical explanation of viscosity: (a) distribution of the thermal velocity and (b) calculation of the upward momentum flux.

With reference to Fig. 2.7 (b), the momentum flux due to the migration of these upward moving molecules, p_{xy}^+ , in a time interval dt is

$$p_{xy}^+ dS dt = \frac{|\mathbf{u}_{th}|}{2} dt dS \frac{\rho}{2} [u(y - \lambda) - u(y)] \approx -\frac{|\mathbf{u}_{th}|}{2} dt dS \frac{\rho}{2} \frac{du}{dy} \lambda, \quad (2.58)$$

where $\frac{\rho}{2}$ has been assumed, accounting for the fact that only half molecules move upward. It is thus possible to conclude that

$$p_{xy}^+ \approx -\frac{|\mathbf{u}_{th}|}{4} \rho \frac{du}{dy} \lambda. \quad (2.59)$$

Analogously, a momentum flux through the surface dS is present due to the downward motion of molecules from the fluid layer above toward the one below. This momentum flux can be evaluated similarly to the previous one as

$$p_{xy}^- \approx \frac{|\mathbf{u}_{th}|}{4} \rho \frac{du}{dy} \lambda. \quad (2.60)$$

Summing the two contributions and equating the macroscopic to the micromechanical momentum flux, we obtain

$$\tau_{xy} = p_{xy}^- - p_{xy}^+ \approx \frac{|\mathbf{u}_{th}|}{2} \rho \frac{du}{dy} \lambda, \quad (2.61)$$

Chapter 3

Bluff-body aerodynamics

...The applicability of governing equations is quite limited without computational tools. In this chapter, we focus on the emergent behaviour of fluid flows around bluff bodies, aiming to develop an intuitive understanding of their structure and organisation. Once again, dimensional analysis proves invaluable, providing the key to generalising experimental observations....

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Chapter 4

Finite differences method

...We now possess an intuition regarding the motion of fluids around bluff bodies. In this chapter, we return to the fluid flow governing equations, aiming to introduce the tools required for their numerical solution. We begin with a one-dimensional problem, whose simplicity allows to clearly highlight the profound implications of numerical approximations...

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4.1 Advection-diffusion equation

We have already derived the Navier-Stokes equations and described some recurrent patterns that emerge in fluid flows, like the spontaneous production of turbulence and the overall flow arrangement established around bluff bodies. We now focus on techniques that allow us to approximate the solutions of these equations using numerical methods.

Unfortunately, the full Navier-Stokes system is far too complex to serve as an introductory model. Therefore, we first consider a simpler problem that still shares many features with the original one: the one-dimensional linear *advection-diffusion* equation.

Chapter 5

Finite volume method basics

...Passing from the one-dimensional case to more general settings requires some effort. In this chapter, the finite volume method is extended to higher-dimensional spaces, paving the way for its application to more challenging problems...

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5.1 Approximations

We already introduced the Finite Volume, FV, method for a one-dimensional domain, showing that it is in fact equivalent to the finite differences formulation in that case. We now proceed to extend the method to multi-dimensional domains, where the advantages provided by FV in dealing with arbitrarily shaped boundaries become apparent.

For simplicity, we consider a two-dimensional domain. It is, however, worth noting that the outlined approach is general and can be easily extended to fully three-dimensional cases.

Chapter 6

Turbulence modelling

...The multi-scale nature of turbulence severely limits our ability to simulate the flow field with the tools we have already developed. We thus venture into the rabbit hole of turbulence modelling. We begin with very little at our disposal, yet standing on the shoulders of giants, we navigate through the night...

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6.1 The need for turbulence modelling

In previous chapters, we derived the fluid flow governing equations and presented the main features which characterise turbulence. Summarising,

Chapter 7

Oscillators and aeroelasticity

...We are now ready to begin investigating the effects of wind on structures. We first revisit the simple oscillator and then explore the behaviour of structures immersed in a smooth flow. The energy transfer between the fluid and the structure may follow unexpected paths, eventually compromising structural stability...

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7.1 Simple oscillator

We now consider the effects of wind on structures, limiting to cases which do not depend on gusts in the incoming wind, a topic which will be considered in due time (see Chapter 9). To keep the discussion self-contained, we first provide a brief summary on complex numbers, rotating vectors and

Chapter 8

Finite volume method complements

...We have now reached a convenient point in our journey, and it is necessary to address a few additional aspects regarding boundary conditions and numerical approximations that were postponed to avoid delaying the discussion in the previous chapter. Some of these aspects might at first appear to be mere details, but they actually play a crucial role in numerical simulations...

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Chapter 9

Wind loading assessment

...We have already analysed the effects of smooth flows on simple immersed vibrating bodies. It is now time to consider more complex structures and turbulent flows, such as wind in the atmospheric boundary layer. A few representative cases are particularly instructive, as they provide insight into the key processes at play...

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9.1 Wind load analysis

We begin by reconsidering the Davenport's wind load chain, reported again in Fig. 9.1 and already introduced in Chapter 1. As already clarified, each

Chapter 10

Bridge aerodynamics

...Among all man-made structures, long-span bridges are perhaps the most evocative due to their slenderness and lightness. These same qualities, however, make them particularly susceptible to even small energy transfers from the surrounding fluid, which can sometimes lead to catastrophic effects. Uncovering these energy transfer pathways requires building on the concepts introduced earlier, deepening our understanding of fluid-structure interaction mechanisms...

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10.1 Historical evolution

We now analyse long-span bridges, which are among the lightest and most slender structures, notoriously sensitive to the wind action. However, before proceeding, it is highly useful to provide a brief description of their historical evolution.

Already at the end of the 19th century, some remarkably long bridges were built, with main spans in the order of 500 m. An iconic example of this period is represented by the Brooklyn Bridge (main span 486 m),

Chapter 11

Application examples

...We have finally reached the end of our journey. Here, we present a few selected examples to bridge ideas with practice, reorganising the content of previous chapters around applications rather than the sequence of exposition. This also allows us to introduce a few additional concepts that, instead of merely concluding the discussion, open the way to new perspectives. These, however, lie beyond the scope of this book and are left for the interested reader to explore...

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11.1 Structure of the examples

We now present a selection of applications, serving a dual purpose. On the one hand, it provides concreteness to the exposition, allowing us to revisit many ideas introduced earlier, highlighting their interaction and practical relevance. On the other, it offers an opportunity to introduce a few addi-

...This example brings our journey to an end. At times, we did not hesitate to take shortcuts, avoiding to engage with certain small details that have been the lifelong headache of numerous researchers and the altar at which countless doctoral students have voluntarily immolated. It is our hope, however, that this discussion has provided the essential concepts needed to engage with more in-depth readings, several of which can be found in the references.